Supplementary Material
MMSE Pair Scheduling: Computational Complexity and Certified Power Allocation

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Sections S1–S3 give the channel construction, curvature coefficients and relaxation examples behind the structural results in the main article. Sections S4–S8 retain the development studies and earlier four-AP experiment; these use different inputs and protocols from the one-AP evaluation in the main figures. Section S9 expands the related-work discussion, and Section S10 explains reproduction from saved data. For the current article, go directly to the full proofs (S11), implemented methods (S12), original and corrective protocols (S13), one-AP outcomes (S14), and provenance, inference and notation (S15). The constructed hardness channels and the ray-traced evaluation channels are distinct input families.

S1 Rational representation used in Theorem 1

Theorem 3.16 of the AIM Minimum Rank Group [1], p. 1639, gives a faithful orthogonal representation of the complement of a tree in $\mathbb{R}^3$, with distinct representing vectors linearly independent. Its leaf induction establishes the geometric antecedent. Here we need the additional requirements of rational unit vectors and a fixed alphabet for a bounded family of forests. The following construction provides those requirements explicitly.

Root each component of a forest with at most seven roles and visit parents before children. The first root receives $\mathbf{e}_3$. A subsequent component root is chosen from

$$\mathbf{r}(t) = \frac{(2t, 2t^2, 1 - t^2 - t^4)^T}{1 + t^2 + t^4}, \qquad t \in \{0, \ldots, 100\}. \tag{A1}$$

These are rational unit vectors. For any previous nonzero vector $\mathbf{w}$, the numerator of $\mathbf{w}^T\mathbf{r}(t)$ is a nonzero polynomial of degree at most four. Orthogonality excludes at most four values of t. Parallelism of unit vectors is equivalent to squared inner product one; after clearing denominators this excludes at most eight values. That polynomial cannot be identically zero because the nonconstant continuous curve would otherwise lie in $\{\mathbf{w}, -\mathbf{w}\}$. At most six preceding vectors forbid at most 72 real values, so an allowed integer exists.

For a child with parent vector $\mathbf{p}$, use $\mathbf{Q} = \mathbf{I}$ if $\mathbf{p} = \mathbf{e}_3$; otherwise let

$$\mathbf{Q} = \mathbf{I} - \frac{(\mathbf{e}_3 - \mathbf{p})(\mathbf{e}_3 - \mathbf{p})^T}{1 - p_3}. \tag{A2}$$

This rational Householder matrix is orthogonal and maps $\mathbf{e}_3$ to $\mathbf{p}$. Choose the child's vector from

$$\mathbf{c}_{\mathbf{p}}(t) = \mathbf{Q}\left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2}, 0\right)^T. \tag{A3}$$

It is unit and perpendicular to its parent. Previous nonparent vectors are nonparallel to $\mathbf{p}$, so undesired orthogonality excludes at most two values and parallelism at most four. An integer in the same candidate set remains available. In parent-before-child order, the parent is the only previous neighbor. Induction therefore realizes exactly the forest's zero-inner-product graph while keeping distinct roles nonparallel.

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For a precise fixed alphabet, enumerate all labelled forests on $\{1, \dots, m\}$ for $1 \le m \le 7$. Root each component at its smallest label, order components by their roots, and traverse each in depth-first order with children in increasing label order. At each choice take the least admissible $t \in \{0, \dots, 100\}$. Define $\mathcal{A}$ as the union of all vectors output by (A1)–(A3), including the initial $\mathbf{e}_3$, over this finite enumeration. This specifies the alphabet exactly without an instance-dependent choice. Its cardinality and rational bit lengths are constants independent of K; a lexicographic ordering of its rational coordinates fixes the input indices. The construction is allowed to use the general forest family, although the reduction needs only forests with at most three edges and one outsider role.

S2 Coefficients for the curvature proof

For $N = \sum_{i,j} c_{ij} x^i y^j$ in (D5), Section S11, use $c_{ij} = c_{ji}$. The nonzero coefficients for $i \ge j$ are

(i, j)	c_{ij}
$4, 4$	a^4
$4, 3$	$4a^4$
$4, 2$	$2a^2(a^2 + 4a - 2)$
$4, 1$	$4a(a^2 + a - 1)$
$4, 0$	$2a^2 - 1$
$3, 3$	$4a^2(4a^2 - a + 1)$
$3, 2$	$4a(2a^3 + 6a^2 - 3a + 1)$
$3, 1$	$4a(3a^2 + 3a - 2)$
$3, 0$	$4(a^2 + a - 1)$
$2, 2$	$2(2a^4 + 16a^3 - a^2 + 1)$
$2, 1$	$4(a + 1)(2a^2 + 2a - 1)$
$2, 0$	$2(-a^2 + 8a - 4)$
$1, 1$	$4(7a - 3)$
$1, 0$	$4(2 - a)(2a - 1)$
$0, 0$	$(3 - 2a)(2a - 1)$

For $a \in [1/\sqrt{2}, 1]$, $a^2 + a - 1 \ge a - 1/2 > 0$, $a^2 + 4a - 2 \ge 4a - 3/2 > 0$, $4a^2 - a + 1 > 0$, $6a^2 - 3a \ge 0$, $3a^2 + 3a - 2 \ge 3a - 1/2 > 0$, and $-a^2 + 8a - 4 \ge 8/\sqrt{2} - 5 > 0$. The remaining terms are immediately nonnegative, while the constant coefficient is strictly positive. This verifies the determinant claim without a sampled Hessian test.

S3 Two distinct obstructions to exact LP recovery

The following examples explain why a valid upper bound need not be attained by one feasible physical schedule. They isolate assignment mixing and power mixing, respectively; neither example is a ray-tracing sample. General-graph matching and frequency-sharing convexification provide the relevant background [2, 3].

S3.1 Assignment mixing with channel-realizable weights

Take six users, three resource-invariant receive dimensions and unit powers. The first three channels are the standard basis of $\mathbb{R}^3$; the other three are the columns of

$$Q = \frac{1}{7} \begin{pmatrix} 6 & -2 & -3 \\ -2 & 3 & -6 \\ -3 & -6 & -2 \end{pmatrix}.$$

Writing $v = (1, 2, 3)^T$ gives $Q = I - vv^T/7$ and $Q^T Q = I$. Each group of three is internally orthogonal, whereas all cross-group correlations are nonzero. Their smallest squared correlation is $4/49$. At unit normalized powers the pair weight is

$$w(r) = \ln \frac{(4 - r)^2}{4}.$$

Every perfect matching has either one or three cross-group edges. A matching containing one cross edge with $r = 4/49$ and two within-group edges is optimal and has value

$$V_{\mathrm{IP}} = 2\ln 4 + \ln \frac{9216}{2401}.$$

There are six within-group edges, three in each triangle. Assign configuration mass $1/6$ to each such edge on each of the three resources. Every user then has total incidence one and every resource has total mass one. This LP solution has value $3\ln 4$, which is optimal because no pair exceeds $\ln 4$. Consequently,

$$V_{\mathrm{LP}} - V_{\mathrm{IP}} = \ln \frac{2401}{2304} = 2\ln \frac{49}{48} > 0.$$

This is a finite-grid example with a singleton unit-power grid and total budget six. It also explains the failure of an unrestricted permutation assignment to enforce disjoint pairs: the two directed three-cycles attain twice the fractional value but are not a pairing. The homogeneous integer problem is still a polynomially solvable general-graph matching problem. The example does not concern formulations that intentionally permit cyclic reuse of users.

S3.2 Power mixing after the pair is already fixed

Take one resource, two unit-gain collinear channels, powers in $[1, 2]$, and total budget $11/5$. Here

$$q(x, y) = \ln \frac{(1 + x + y)^2}{(1 + x)(1 + y)}.$$

Both partial derivatives are nonnegative on this box: for example, $q_x = (1 + x - y)/[(1 + x + y)(1 + x)]$. An optimum therefore exists on $x + y = 11/5$. On this line the numerator is constant and the denominator is concave in x; its minimum is at $x = 1$ or $x = 6/5$. Hence the integral continuous optimum is $\ln(128/55)$.

The continuous configuration LP can mix weight $4/5$ at $(1, 1)$ and $1/5$ at $(1, 2)$. Its average power is $11/5$ and its value exceeds the deterministic optimum because

$$\left(\frac{9}{4}\right)^4 \frac{8}{3} > \left(\frac{128}{55}\right)^5.$$

The decisive comparison uses rational arithmetic; no rounded logarithm is needed. The pairing is already fixed, so branching only on assignment variables cannot resolve this gap. A single-slot interpretation requires optimization of actual powers, whereas a time-sharing interpretation would describe a different feasible service model.

S4 Development studies with imperfect channel information

These studies explain the scope of the final optimization contribution. They were completed during method development and are not independent confirmatory replications. Their overlapping networks, policies and noise realizations must not be counted as independent samples. In particular, a nominal-CSI optimization certificate does not certify a rate under channel-estimation error.

S4.1 Pilot stopping and conditional reliability

Two bounded development rounds constructed 1,800 channel/noise paths and 5,400 policy-evaluation rows. A split-conformal calibration of the maximum prefix error provides a simultaneous event under exchangeability and fixed pre-calibration models. Its coverage is marginal; it is not conditional coverage among the paths on which the stopping rule activates, nor a guarantee under distribution shift.

The learned stopping policy did not outperform the best development-selected fixed pilot length in any of the nine IID noise/coherence settings. The latter is an exploratory comparator selected after observing development data, not a deployable rule selected independently. In the second round at noise variance one

and coherence length 200, exactly one of 80 paths stopped early, and that path violated the sufficient oracle comparison used by the certificate. Thus the marginal count was 1/80, whereas the count conditional on early stopping was 1/1. Neither observation supports a statement that 95% of the observed stops were safe. This result led us to stop further tuning of the early-stopping approach.

S4.2 Equal-cost selective acquisition

Four development contexts each contained eight paired networks. The joint, random and weakest-user acquisition rules each used five pilot symbols and total normalized pilot energy 17, followed by the same six-schedule adapted I-WMMSE optimization with power budget 0.4 W and a five-second external limit. All 192 information jobs completed. The results below include pilot overhead and normalize by the two active resources; computation airtime is not subtracted. Intervals are the recorded descriptive 95% development intervals, not multiplicity-adjusted confirmatory tests.

Geometry	Noise	Joint minus random	Joint minus weakest
Uniform	0.1	-0.022050 $[-0.059435, 0.001975]$	-0.007660 $[-0.031605, 0.015987]$
Uniform	1.0	0.039351 $[-0.053686, 0.168140]$	-0.002897 $[-0.049399, 0.045508]$
Nearby	0.1	0.008837 $[-0.031740, 0.051078]$	-0.002970 $[-0.036133, 0.032290]$
Nearby	1.0	0.006285 $[-0.024990, 0.035534]$	-0.013387 $[-0.059055, 0.025706]$

All intervals span zero, and all joint-minus-weakest means are negative. This does not prove equivalence or impossibility. It supplies no evidence of a reliable selective-acquisition improvement in the tested family. An auxiliary uniform-probe arm required separately observable user statistics without an established equal-resource physical multiplexing implementation; it is therefore excluded from the primary equal-cost comparison.

Certificate-based early termination activated on zero of the 32 networks. Joint and joint-with-certificate returned identical achieved rates and evaluated all six schedules. Observed timing variation cannot be attributed to skipped optimization work. The bound remained a nominal continuous-power bound, not a channel-error guarantee.

S4.3 A finite-grid opportunity check

For the same 32 development cases, exhaustive enumeration of six schedules and the prescribed finite power grid fixed the initial receiver estimate while granting the action selector true-channel knowledge. The aggregate positive-part net opportunity was 0.5317% of the no-probe rate, below the prospectively declared 1% continuation threshold. In 23 of 32 cases even the best such action could not repay the extra pilot symbol. The ordinary grid-LP opportunity bound was 0.5918%, illustrating separate relaxation looseness. This is a pointwise finite-family diagnostic aggregated over fixed cases, not a population impossibility theorem or a bound for arbitrary continuous-power and receiver-information methods. It motivated stopping further selector tuning and concentrating on optimization structure.

S5 CPU/GPU engineering study and minimum-power evidence

This ancillary study used one existing development instance at each of 8, 16 and 32 users. It measures an implemented backend; it does not establish a GPU scheduling advantage. Four initial GPU attempts reached their 40-second external limits and remain recorded failures. A separate diagnostic isolated 51.765 seconds in the first sparse matrix–vector product with an unsorted, noncanonical CSR matrix. Canonicalizing the host CSR representation before transfer preserved every matrix entry and its nonzero count. The diagnostic did not separately identify compilation and sorting costs inside the library call.

S5.1 Coefficient construction and LP optimization

Five alternating CPU/GPU coefficient-construction repetitions used FP64 and all combinations of pairs, resources and nine power profiles. Medians below are milliseconds. These repetitions describe implementation variability on one input per size, not variation across independent channel scenes.

Users	Columns	CPU	GPU including transfers	Maximum coefficient error
8	1,008	0.1542	1.8749	1.42×10^{-14}
16	8,640	0.4543	1.6412	1.08×10^{-13}
32	71,424	3.0479	1.8857	6.04×10^{-14}

The error unit is summed-resource bit/s/Hz. GPU coefficient construction including transfers was faster only at the largest tested size. Whole optimization costs give a different result:

Users	CPU LP solve (s)	GPU worker (s)	Last-iterate violation	GPU result
8	0.01030	32.0370	0	Tolerance met
16	0.08465	31.8222	0.0136358	Time limit; infeasible
32	2.95306	31.8401	0.1405925	Time limit; infeasible

CPU whole-worker times were 0.628, 0.573 and 3.476 seconds. No GPU LP speedup follows. Only the last eight-user GPU iterate was feasible at the declared numerical tolerance; its running average was not. Both saved candidates were infeasible at 16 and 32 users, so their objectives are not feasible rates or primal lower bounds. Saved numerical dual vectors were checked with an explicit floating-point margin, not exact directed arithmetic; they bound only the finite-grid model under their stated numerical qualification. They do not replace the continuous-power certificates in the main paper.

Final peak process working sets were approximately 1.657, 1.659 and 1.676 GB. CuPy managed-pool totals were 226,816, 1,906,688 and 14,610,944 bytes; these are not measurements of all driver allocations or whole-device peak memory. The environment was Python 3.10.19, NumPy 2.1.3, SciPy 1.15.3, CuPy 13.6.0 and an NVIDIA GeForce RTX 5060 Laptop GPU. The driver query reported 610.74 and the runtime reported CUDA 12.4. Existing compiled caches were retained, so these are not pristine cold-install measurements.

S5.2 Minimum grid power is a different endpoint

Historical grid-ILP selections were revalidated at targets equal to 95% of their fixed-power incumbent throughput. The old LP values are numerical lower bounds for that finite-grid minimum-power problem, not new exact continuous minimum-power certificates.

Users	Target (Mbps)	Achieved (Mbps)	Grid power (W)	Numerical LP (W)
8	123.766432	124.286608	0.60	0.5826457982
16	240.980704	242.176902	1.20	1.1583640115
32	442.397913	443.393155	2.35	2.3114858403

Occupied bandwidth differs across these legacy sizes. These are reduced-throughput targets, so the power reductions are not equal-throughput gains. A separate fixed-schedule continuous experiment already achieved the 95%-rate target with uniform scaling at 0.560795, 1.120640 and 2.262241 W. Its schedules and target values differ from the rescheduled grid study; subtracting these numbers would not constitute a matched superiority comparison. Under maximum-rate fixed-schedule comparisons, SLSQP and adapted I-WMMSE improved essentially negligibly over uniform power on those inputs. These observations concern the stated inputs and endpoints; they do not establish a general advantage for either power-allocation method.

S6 Prospective fixed-estimator CSI follow-up

S6.1 Fixed estimators and scheduling decisions

The separate CSI follow-up used 120 four-user networks drawn from O1_60 rows 101–200, with four distinct users per network and replacement across networks. All 18,100 coordinates in that pool were checked against the 18,100 old row-1–100 coordinates; none overlapped. The location, observation-noise and bootstrap seeds were 20261101, 20261102 and 20261103, respectively. Repeated sampling from one fixed map supports inference conditional on that map, not inference over independent propagation scenes. No sampled channel was removed; no sampled user had zero channel norm.

Four coherent APs with eight antennas each and subcarriers 0 and 16 form each 64-complex-coordinate user channel. This ancillary panel preserves the original statistical normalization: train-only RMS scaling, unit user powers and noise variance 0.1. These are normalized units, distinct from the physical-watt/noise-figure optimization experiment. DFT pilots have length $\tau \in \{4, 8\}$ and unit per-symbol power. Every estimator at the same τ observes the same noisy pilot realization. The training mean and covariance, CNN, GAN-augmented CNN and categorical imitation model were fixed before this follow-up; no retraining or model selection used the new pool.

The historical training split contained 200 four-user networks, with separate 40-network validation, calibration and test splits. The residual CNN operates on 64 real/imaginary coordinate pairs, using two 16-channel, width-three Conv1D/ReLU layers followed by a two-channel width-one correction added to the input. Its training used Adam with learning rate 10^{-3}, squared error, batch size 32 and 30 epochs. The train-only GAN maps a 16-dimensional latent input through dense layers of widths 64 and 128; its discriminator uses widths 64 and one. The GAN uses Adam at 2×10^{-4}. Generated training channels are paired with newly simulated observations. CNN and augmented-CNN training used equal final-estimator example and update counts, but GAN training adds computation. The stored weights define the implementation evaluated here; the experiment does not reproduce published curves from another study. The imitation model maps 512 real features through 64 ReLU units to six logits, trained with categorical cross-entropy for 30 epochs.

Each estimated channel selects a no-SIC MMSE receiver and a schedule; actual channels evaluate signal and interference. All four users are served once on two resources. There are exactly six labelled pair schedules, so enumeration supplies the discrete optimum of the declared decision problem. A per-cell LP is recorded separately. The imitation and fixed-schedule controls use the same CNN observation stream at $\tau = 4$. Reported net spectral efficiency is

$$R_{\text{net}} = \frac{200 - \tau}{200} \frac{1}{2} \sum_{i=1}^{4} \log_2(1 + \gamma_i^{\text{actual}}),$$

which includes pilot overhead and normalizes by the two active resources. It excludes blocking computation/control time, decoding failures and circuit energy. The eight-symbol arm spends twice the pilot energy of the four-symbol arm; cross-length comparisons are not equal-energy comparisons.

S6.2 Complete prospective outcomes

All 1,440 planned method/length/network cells completed successfully. The prespecified primary outcome was the paired CNN-minus-LMMSE mean at $\tau = 4$. It was -0.023798 bit/s/Hz, with sample standard deviation 0.143477 and Student-t_{119} 95% interval $[-0.049732, 0.002137]$. The 10,000-resample paired bootstrap sensitivity interval was $[-0.049089, 0.002429]$. The primary interval includes zero, so CNN superiority is not established. This is not a proof of equivalence.

	$\tau = 4$			$\tau = 8$		
Method	Net SE	NMSE	Min-user SE	Net SE	NMSE	Min-user SE
LS	16.420464	0.015243	3.648801	16.431631	0.007613	3.698542
LMMSE	16.513690	0.004643	3.672461	16.458718	0.002787	3.713057
CNN	16.489893	0.007274	3.666157	16.442293	0.003766	3.704346
GAN-CNN	16.459948	0.008259	3.654948	16.432855	0.004137	3.703641
True-CSI control	17.145864	0	3.876895	16.795949	0	3.797775
Imitation/CNN	15.626223	0.007274	3.421179	—	—	—
Fixed/CNN	15.457598	0.007274	3.380715	—	—	—

Every displayed mean uses all 120 networks. SE quantities are bit/s/Hz under the preceding normalization; NMSE is dimensionless. The true-CSI control is restricted to this receiver and finite scheduling family. The secondary values are descriptive and were not used to replace the prespecified endpoint or choose a favorable estimator.

Parent-measured generation and evaluation times were 18.625 and 10.156 seconds, both below their 600-second caps. The complete CSI parent, including setup and verification, took 39.646 seconds with

peak process-tree RSS approximately 574 MB. These are host measurements of this run, not wireless service latency. All models and source manifests remained fixed. This follow-up supports retaining the imperfect-CSI findings as ancillary evidence; it supplies no new channel-estimation advantage for the nominal optimization method.

S7 Historical four-AP experimental procedures

The earlier experiment used four coherent APs and comprised 176 method executions. Its procedures and complete outcomes are reported here and in Section S8. The one-AP studies used in the main figures are described separately in Sections S12–S14. The numerical master and the continuous certificate have different roles. SciPy/HiGHS solves the finite-grid MILP and LP using floating-point coefficients. An incumbent is converted to rational powers, clipped to the declared box, and any positive budget excess is removed in user-index order without crossing a lower bound. The resulting integral schedule is checked against every user and resource equality. Its objective is evaluated with outward logarithmic bounds. A solver status or a small numerical master gap is not itself a certificate for continuous powers.

The master uses all $S\binom{K}{2}|\mathcal{G}|^2$ configurations. These counts are 1008 for $K = 8$, $S = 4$ and three power levels, 5488 for the seven-level control, and 8640 for $K = 16$, $S = 8$ and three levels. Each global eight-user arm reserves at most 8 seconds for its fresh grid computations, with nominal native MILP and LP caps of 4 seconds each. Remaining charged time includes startup and all other work; these caps do not grant additional time. The sixteen-user grid-plus-pricing arm allows at most 20 seconds for the grid stage, with 10-second native caps. The standalone grid arm uses its remaining budget for the same two numerical problems. Early termination and incumbent availability are retained separately from optimal-return status.

The continuous repair prices all 112 pair/resource rectangles at $K = 8$ and all 960 at $K = 16$. Saved channels use complex64 storage. Their real and imaginary components are promoted without changing their values to Python floating-point scalars, then converted to exact rational numbers. The certificate concerns these stored values, not an unquantized propagation field. SymPy isolates the real roots of the resulting rational polynomials. At refinement digits 12, 24 and 36, candidate intervals are intersected with their rectangles and evaluated using rational interval operations. Logarithms use Decimal precision 60: rational-to-decimal conversion is bracketed, and correctly rounded logarithms are expanded to adjacent representable values. The requested pair-price enclosure width is 10^{-7} nat. Failure to resolve an enclosure is reported; no floating root is substituted. The price upper endpoint, rather than a midpoint estimate, enters the repaired dual. A partial price list never replaces the universal valid upper bound.

Both local arms traverse the same deterministic stream of assignments and start each schedule at the lower-power vector and the feasible uniform vector. SLSQP uses analytic gradients, at most 500 iterations per start and a 10^{-10} stopping tolerance. Adapted I-WMMSE alternates the exact two-user receiver, inverse-MSE weights and clipped transmit amplitudes. A common nonnegative power multiplier is bracketed by at most 100 doublings and refined with 60 bisections when needed. It uses at most 500 iterations and a relative objective-improvement stopping tolerance of 10^{-9}; objective descent above $10^{-7}\max(1, |q|)$ is recorded as a numerical failure. A feasible incumbent remains usable when local stationarity is not established. Enumeration stopping early changes the number of schedules examined, not the declared feasible set. These adaptations do not reproduce the original authors' full network scenarios.

S7.1 Seeds, retention and scope

The nominal user-location seed is 20261111, the within-network method-order seed is 20261112, the descriptive paired-bootstrap seed is 20261113, and the sixteen-user location seed is 20261114. The first four nominal network IDs, selected before outcomes, form the grid control. Its grids are $\{0.05, 0.10, 0.20\}$ W and $\{0.05, 0.075, 0.10, 0.125, 0.15, 0.175, 0.20\}$ W. Their charged budget is 29 seconds, with a 30-second external watchdog. All 144 eight-user, eight grid-control and 24 sixteen-user cells are retained, including unsuccessful process exits. The records distinguish an undispatched cell from a dispatched failure; no cell was ultimately omitted. No cell is repeated to replace an unfavorable or failed outcome.

Channels retain the generator's finite-bandwidth path-delay treatment. The generator reports paths extending beyond the useful OFDM duration; its default handling and the original warnings are preserved

in the logs. The experiment does not validate a cyclic-prefix, mobility, Doppler, fronthaul or hardware-impairment model. Repeated user locations across independently sampled networks are permitted by the conditional sampling law. These networks are replications conditional on one ray-traced scene, not independent deployment scenes.

S7.2 Time limits, failures and endpoint ambiguity

Before launch, each cell writes a durable dispatch marker. A parent process validates a candidate, writes an immutable candidate file, and records a hash-bound receipt with its persistence time. Only a candidate persisted by the scientific deadline can improve the reported interval. A later worker snapshot or a favorable late return cannot qualify retrospectively. A universal interference-free ceiling and a feasible uniform schedule provide initial fallback bounds. External interruption reconciles every planned cell; a dispatched process without usable output is distinguished from a cell never launched.

The original protocol's general sentence treating all dispatched failures as unsuccessful conflicts with the more specific pre-outcome rule preserving timely verified bounds and the prespecified primary-analysis function. The operational primary retains a valid timely certificate even if its process subsequently fails. The failure remains a failure of normal process completion. This conflict was documented after the first observed storage error, without changing the prespecified protocol, method, trial, or primary function. Accordingly, we also report a clearly post-hoc normal-return sensitivity: success requires both a timely certificate and process status `RETURNED`. This identical rule is applied to every network and both primary arms. No extra elapsed-time condition or favorable exclusion is introduced. Internal worker errors retained behind a normal process return are listed separately. The sensitivity does not replace the primary or create a second opportunity to select a positive conclusion.

The logs establish Windows `Access denied` errors during replacement of a mutable snapshot file; their underlying operating-system cause has not been established. The first affected full-method cell, B-00, has an accepted receipt at 9.4834 seconds with gap 0.00928254 nat and a subsequent process error at 9.5912 seconds. A valid earlier interval and a later operational failure can coexist. Final process duration is not used as time to first certification. No runtime repair or rerun followed these errors.

S8 Historical four-AP numerical results

This section reports all 176 executions of the four-AP study in Section S7. These results are separate from the one-AP evaluation in the main figures and are not pooled with it. All entries below are the final timely eligible intervals, not later worker snapshots. Q is the sum of per-resource spectral efficiencies in bit/s/Hz; G is the global interval width in natural-log units; t is total parent process duration in seconds, not time to first certification. R denotes normal process return and E an unsuccessful process exit. A star marks an internal error retained in the eligible snapshot, even when the process returned normally. Every planned cell has a valid fallback or improved interval; no cell was administratively omitted. The detailed CSV additionally records radiated power, minimum-user SE, fairness, memory, pricing and coverage fields. The prespecified primary contingency table, ordered (full success, generic success), is $n_{00} = 19$, $n_{01} = 0$, $n_{10} = 5$, $n_{11} = 0$. Its conservative 95% interval is $[-0.1070655, 0.4506892]$. The separately disclosed normal-return sensitivity gives $n_{00} = 21$, $n_{01} = 0$, $n_{10} = 3$, $n_{11} = 0$ and interval $[-0.1464695, 0.3523712]$. Neither establishes superiority. The primary bootstrap is descriptive and does not supersede the conservative interval.

S8.1 Eight-user panel: all 144 cells

ID	Arm	Q	G	t	Certificate	Status
B-00	Generic	98.96890	0.50540	19.706	0	R
B-00	Curvature	98.96890	0.00933	5.053	1	R
B-00	Full	98.96890	0.00928	9.591	1	E
B-00	Grid	98.96890	7.03664	0.899	0	R
B-00	SLSQP	98.96890	7.03664	9.740	0	R
B-00	I-WMMSE	97.11448	8.32203	19.673	0	R
B-01	Generic	97.17456	0.37511	19.700	0	R

ID	Arm	Q	G	t	Certificate	Status
B-01	Curvature	97.17456	0.00044	1.506	1	R
B-01	Full	97.17456	0.00044	5.982	1	R
B-01	Grid	97.17456	7.89643	0.869	0	R
B-01	SLSQP	97.17456	7.89643	9.897	0	R
B-01	I-WMMSE	94.89364	9.47744	19.680	0	R
B-02	Generic	99.61673	0.40727	19.712	0	R
B-02	Curvature	99.61673	0.01348	19.685	0	R
B-02	Full	99.61673	0.01323	19.716	0	R
B-02	Grid	99.61673	6.97465	0.905	0	R
B-02	SLSQP	99.61673	6.97465	9.910	0	R
B-02	I-WMMSE	99.61673	6.97465	17.058	0	R*
B-03	Generic	97.55323	0.37490	19.683	0	R*
B-03	Curvature	97.55323	0.02018	19.694	0	R
B-03	Full	97.55323	0.02455	11.549	0	R*
B-03	Grid	97.55323	7.30056	0.847	0	R
B-03	SLSQP	97.55323	7.30056	9.835	0	R
B-03	I-WMMSE	97.55323	7.30056	19.692	0	R
B-04	Generic	97.27705	0.41517	19.702	0	R
B-04	Curvature	97.27705	0.00360	13.582	1	R
B-04	Full	97.27705	0.00360	17.879	1	R
B-04	Grid	97.27705	7.83775	0.867	0	R
B-04	SLSQP	97.27705	7.83775	9.793	0	R
B-04	I-WMMSE	97.27705	7.83775	19.690	0	R
B-05	Generic	101.02330	0.43283	19.693	0	R
B-05	Curvature	101.02330	0.14781	19.707	0	R
B-05	Full	101.02330	0.15992	19.717	0	R
B-05	Grid	101.02330	7.63156	0.879	0	R
B-05	SLSQP	101.02330	7.63156	9.751	0	R
B-05	I-WMMSE	99.12979	8.94404	15.550	0	R*
B-06	Generic	96.49894	0.44244	19.700	0	R
B-06	Curvature	96.49894	0.00258	6.562	1	R
B-06	Full	96.49894	0.02000	19.721	0	R
B-06	Grid	96.49894	7.08581	0.877	0	R
B-06	SLSQP	96.49894	7.08581	9.735	0	E
B-06	I-WMMSE	96.49894	7.08581	19.689	0	R
B-07	Generic	96.34364	1.25823	3.096	0	R*
B-07	Curvature	96.34364	0.24109	19.687	0	R
B-07	Full	96.34364	0.26358	19.732	0	R
B-07	Grid	96.34364	8.66321	0.879	0	R
B-07	SLSQP	96.34364	8.66321	9.916	0	R
B-07	I-WMMSE	95.77378	9.05822	19.690	0	R
B-08	Generic	96.53631	0.49040	19.691	0	R
B-08	Curvature	96.53631	0.36058	7.584	0	R*
B-08	Full	96.53631	0.27587	19.733	0	R
B-08	Grid	96.53631	8.54707	0.871	0	R
B-08	SLSQP	96.53631	8.54707	9.861	0	R
B-08	I-WMMSE	95.27775	9.41944	19.681	0	R
B-09	Generic	98.65888	0.41435	19.699	0	R*
B-09	Curvature	98.65888	0.03599	19.697	0	R
B-09	Full	98.65888	0.03813	19.723	0	R
B-09	Grid	98.65888	7.96941	0.870	0	R
B-09	SLSQP	98.65888	7.96941	9.654	0	R
B-09	I-WMMSE	95.73370	9.99699	19.682	0	R
B-10	Generic	94.31388	0.77052	19.695	0	E
B-10	Curvature	94.31388	0.03178	19.703	0	R
B-10	Full	94.31388	0.05323	19.743	0	R
B-10	Grid	94.31388	8.01906	0.839	0	R
B-10	SLSQP	94.31388	8.01906	9.721	0	R
B-10	I-WMMSE	93.84583	8.34349	19.676	0	R
B-11	Generic	98.13141	0.45400	19.684	0	R
B-11	Curvature	98.13141	0.20963	8.049	0	R*
B-11	Full	98.13141	0.21713	12.124	0	R*
B-11	Grid	98.13141	7.77897	0.845	0	R
B-11	SLSQP	98.13141	7.77897	9.772	0	R
B-11	I-WMMSE	97.34787	8.32208	19.684	0	R
B-12	Generic	97.51865	0.51661	19.711	0	R

ID	Arm	Q	G	t	Certificate	Status
B-12	Curvature	97.51865	0.00858	19.127	1	R
B-12	Full	97.51865	0.01786	19.730	0	R
B-12	Grid	97.51865	7.63473	0.851	0	R
B-12	SLSQP	97.51865	7.63473	9.746	0	R
B-12	I-WMMSE	97.32490	7.76903	19.685	0	R
B-13	Generic	99.48461	0.39396	19.699	0	R
B-13	Curvature	99.48461	0.03737	19.688	0	E
B-13	Full	99.48461	0.03602	19.730	0	R
B-13	Grid	99.48461	7.49787	0.878	0	R
B-13	SLSQP	99.48461	7.49787	9.757	0	R
B-13	I-WMMSE	99.02518	7.81632	19.689	0	E
B-14	Generic	96.24070	0.59486	19.693	0	E
B-14	Curvature	96.24070	0.15337	19.695	0	E
B-14	Full	96.24070	0.16990	19.743	0	R
B-14	Grid	96.24070	7.52042	0.868	0	R
B-14	SLSQP	96.24070	7.52042	9.892	0	R
B-14	I-WMMSE	94.10534	9.00053	19.680	0	R
B-15	Generic	98.49876	0.48559	19.703	0	R
B-15	Curvature	98.49876	0.03572	19.704	0	R
B-15	Full	98.49876	0.03963	19.730	0	E
B-15	Grid	98.49876	6.92559	0.875	0	R
B-15	SLSQP	98.49876	6.92559	9.718	0	R
B-15	I-WMMSE	97.42537	7.66960	19.682	0	R
B-16	Generic	94.31804	0.53979	19.704	0	R
B-16	Curvature	94.31804	0.23873	19.684	0	R
B-16	Full	94.31804	0.26647	19.733	0	R
B-16	Grid	94.31804	8.12470	0.884	0	R
B-16	SLSQP	94.31804	8.12470	9.849	0	R
B-16	I-WMMSE	93.54588	8.65991	19.696	0	R
B-17	Generic	97.55363	0.38597	19.693	0	R
B-17	Curvature	97.55363	0.11399	3.038	0	R*
B-17	Full	97.55363	0.03649	19.718	0	R
B-17	Grid	97.55363	7.22778	0.877	0	R
B-17	SLSQP	97.55363	7.22778	9.829	0	R
B-17	I-WMMSE	96.15016	8.20059	19.693	0	R
B-18	Generic	99.63787	0.53941	19.716	0	E
B-18	Curvature	99.63787	0.05944	19.703	0	R
B-18	Full	99.63787	0.08208	19.721	0	E
B-18	Grid	99.63787	8.12115	0.876	0	R
B-18	SLSQP	99.63787	8.12115	9.712	0	R
B-18	I-WMMSE	99.15712	8.45438	19.693	0	R
B-19	Generic	99.69321	0.38742	19.683	0	R
B-19	Curvature	99.69321	0.00027	1.518	1	R
B-19	Full	99.69321	0.00027	5.981	1	R
B-19	Grid	99.69321	6.81901	0.871	0	R
B-19	SLSQP	99.69321	6.81901	9.723	0	R
B-19	I-WMMSE	97.53465	8.31521	19.692	0	R
B-20	Generic	99.63108	1.00268	2.514	0	R*
B-20	Curvature	99.63108	0.03556	19.690	0	R
B-20	Full	99.63108	0.04039	19.726	0	R
B-20	Grid	99.63108	7.32233	0.876	0	R
B-20	SLSQP	99.63108	7.32233	10.051	0	R
B-20	I-WMMSE	96.85560	9.24615	19.671	0	R
B-21	Generic	98.34496	0.38728	19.692	0	R
B-21	Curvature	98.34496	0.13676	19.702	0	R
B-21	Full	98.34496	0.14813	19.738	0	R
B-21	Grid	98.34496	7.16272	0.866	0	R
B-21	SLSQP	98.34496	7.16272	9.790	0	R
B-21	I-WMMSE	96.87832	8.17932	16.558	0	R*
B-22	Generic	98.01227	0.45398	19.690	0	R
B-22	Curvature	98.01227	0.00185	5.579	1	R
B-22	Full	98.01227	8.35868	0.841	0	R*
B-22	Grid	98.01227	8.35868	0.892	0	R
B-22	SLSQP	98.01227	8.35868	9.717	0	R
B-22	I-WMMSE	97.48417	8.72473	19.670	0	R
B-23	Generic	99.51395	0.44557	19.707	0	R

ID	Arm	Q	G	t	Certificate	Status
B-23	Curvature	99.51395	0.00032	1.482	1	R
B-23	Full	99.51395	0.00032	6.091	1	E
B-23	Grid	99.51395	6.83890	0.887	0	R
B-23	SLSQP	99.51395	6.83890	9.789	0	R
B-23	I-WMMSE	99.15893	7.08498	19.684	0	R

S8.2 Sixteen-user panel: all 24 cells

Pricing completeness is required before a repaired dual is used. Seven grid-plus-pricing cells retain their universal ceiling after an internal storage error; their partial price lists do not imply a continuous dual certificate. The completed price list for scale-07 produces relative gap 0.713123%. Local arms retain the same universal ceiling. All columns include failed internal processing.

ID	Arm	Q	G	t	Priced/960	Status
00	Grid+price	202.33302	12.70820	32.455	796	R*
00	I-WMMSE	190.99969	20.56387	59.693	—	R
00	SLSQP	194.74173	17.97008	59.670	—	R
01	Grid+price	199.23738	13.47549	16.539	389	R*
01	I-WMMSE	187.36647	21.70378	59.690	—	R
01	SLSQP	188.09774	21.19690	59.676	—	R
02	Grid+price	197.43897	13.93191	30.038	734	R*
02	I-WMMSE	187.27627	20.97616	59.683	—	R
02	SLSQP	189.90611	19.15329	59.680	—	R
03	Grid+price	195.70516	13.50835	27.103	664	R*
03	I-WMMSE	182.94596	22.35236	59.686	—	R
03	SLSQP	185.39190	20.65696	59.691	—	R
04	Grid+price	200.60082	13.79894	15.480	367	R*
04	I-WMMSE	185.33931	24.37740	59.674	—	R
04	SLSQP	186.85101	23.32958	12.112	—	R*
05	Grid+price	201.35599	14.05861	26.544	643	R*
05	I-WMMSE	187.23267	23.84815	42.379	—	R*
05	SLSQP	187.23267	23.84815	2.021	—	R*
06	Grid+price	200.24919	14.02126	26.057	638	R*
06	I-WMMSE	188.04976	22.47727	59.670	—	R
06	SLSQP	189.44217	21.51212	59.686	—	R
07	Grid+price	202.93632	1.00311	38.878	960	R
07	I-WMMSE	189.14150	22.71580	32.793	—	R*
07	SLSQP	193.21342	19.89337	59.678	—	R

Fig. S1 shows the relative-gap consequences of these incomplete pricing runs. Only case 07 falls below the descriptive 1% line; the other seven grid-plus-pricing cases retain their universal ceiling.

S8.3 Finite-grid ILP and LP solutions

Both numerical solves returned optimal status (HiGHS status 0) for each of these eight grid-control cells. All returned normally with no retained internal error. Equality after rounding is a numerical finite-grid observation, not a proof for the continuous domain.

ID	Levels	IP (nat)	LP (nat)	Q	t (s)
B-00	3	68.600014	68.600014	98.968900	0.851
B-00	7	68.600014	68.600014	98.968900	0.978
B-01	3	67.356271	67.356271	97.174558	0.849
B-01	7	67.356271	67.356271	97.174558	0.927
B-02	3	69.049057	69.283727	99.616733	0.865
B-02	7	69.049057	69.283727	99.616733	0.950
B-03	3	67.618748	67.618748	97.553232	0.850
B-03	7	67.618748	67.618748	97.553232	0.946

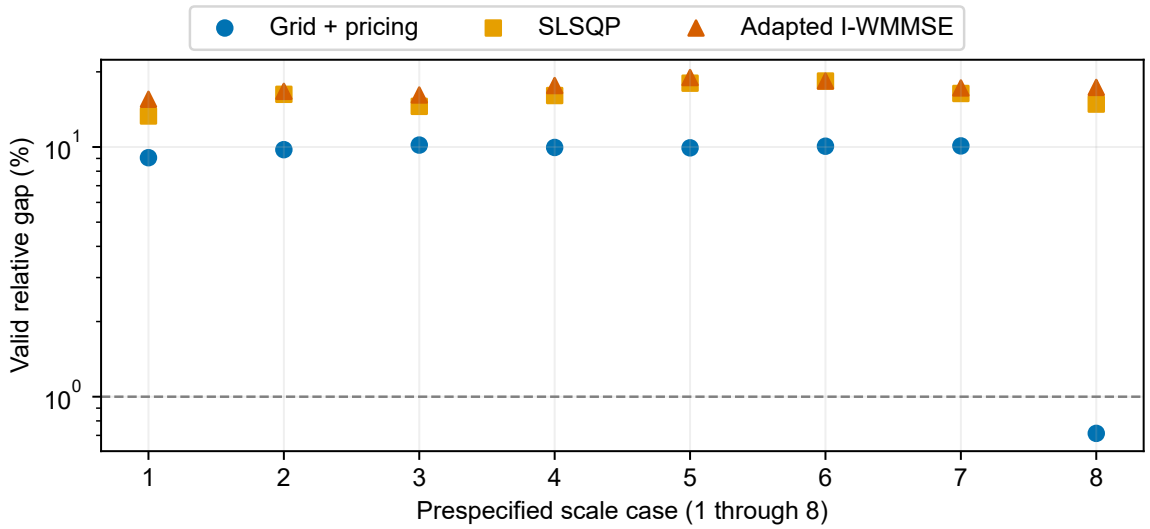


Figure S1: All eight scale cases and all three arms; plotted positions 1–8 correspond to IDs 00–07 in the table. The plotted relative gap is $100(U - L)/L$ percent for $L > 0$, not measured regret. Partial pricing and internal failures are retained. The dashed 1% line is descriptive; no scale cell meets the 0.01-nat criterion.

S8.4 Fairness, power and memory

The following are descriptive means across all 24 eight-user or eight sixteen-user networks. Minimum-user SE is in bit/s/Hz, Jain fairness is dimensionless, and memory is the maximum observed worker RSS in decimal MB. Every eligible schedule is checked against the exact rational power budget; displayed decimal sums can differ at floating-point roundoff. Zero-rate-user counts are zero in all 176 cells. These are not QoS-constrained designs.

Users	Arm	Power (W)	Min-user SE	Jain	Peak RSS (MB)
8	Curvature	0.8000	11.197123	0.997842	126.202
8	Full	0.8000	11.197123	0.997842	134.148
8	Grid	0.8000	11.197123	0.997842	120.119
8	I-WMMSE	0.8000	11.154296	0.997595	106.996
8	Generic	0.8000	11.197123	0.997842	126.861
8	SLSQP	0.8000	11.196738	0.997841	107.958
16	Grid+price	1.6000	11.421602	0.998330	148.922
16	I-WMMSE	1.6000	9.833809	0.995248	108.339
16	SLSQP	1.6000	10.036867	0.995225	109.015

The nominal runner returned after 2899.8007 seconds. Its outer control process subsequently exited unsuccessfully when launcher metadata collided with the already-written primary-analysis filename. The original analysis payload, all 176 records and all 72 startup-source hashes were retained. External reconciliation observed no active timed workers at 3020.8174 seconds after start; this later observation is not an exact controller runtime. Outputs then occupied 402,225,068 bytes. No scientific trial was repeated, and no runtime code was repaired after observing outcomes. The reproducibility package preserves both the failure and recovered closure records.

S9 Technical context and source-specific comparisons

The main article compares the closest studies in a compact table. This section gives the technical distinctions behind that comparison and explains the roles of the remaining sources. The reference model serves every user exactly once, places two users on each orthogonal resource, recomputes linear MMSE reception without

SIC, and optimizes positive box-constrained powers under one total budget. A difference from this model limits direct transfer of a result; it need not be a weakness of the source study.

S9.1 MMSE rates, pairing and frequency allocation

Haustein [4] gives the recomputed-MMSE SINRs and no-SIC sum rate in (4.33)–(4.36), printed p. 48, and the two-user SIMO specialization in (4.47)–(4.49), p. 51. The same discussion identifies general nonconcavity. These receiver identities and the benefit of orthogonal channel vectors are therefore background for the present curvature and pricing results. The later quadratic calculation for SIC with an SNR gap concerns a different objective.

Fan et al. [5], (3)–(12), combine prescribed-power, weighted post-MMSE pair rates with contiguous SC-FDMA resource patterns and binary allocation. Their branch-and-bound algorithm is direct precedent for joint pairing and resource optimization; power control is explicitly outside their optimization. Literal reproduction also requires reconciling their printed exact-cover equations with the description permitting optional service. The present full-service equalities make that convention explicit.

Ruder et al. [6] distinguish homogeneous-resource pairing from frequency-selective grouping and allocation, using capacity and approximate-BER criteria. Section 6.1, p. 97, describes the latter as a three-dimensional assignment problem that is hard in general. This supplies scheduling context, but does not by itself establish hardness for fixed-dimensional channel vectors. The ICC 2011 precursor and the 2013 extension are treated as one work family. Ruder et al. [7], (3)–(6), pp. 90–91, and (26)–(28), p. 94, study fixed-power MMSE pair allocation with channel interpolation and prediction. Their experiment distinguishes imperfect scheduling CSI from perfect payload equalization CSI. Contiguous resource patterns and optional service also differ from one mandatory pair per resource. Our nominal bounds apply to the declared channel realization; they do not bound actual rates after CSI mismatch.

Viterbo and Hottinen [8], Section II, pp. 243–244, formulate equal-power MMSE pairing with channels constant across a frame. Their disjoint-pair service family in (8) and expanded cyclic family in (11) are different feasible sets. An unrestricted permutation need not consist of disjoint two-cycles, but that observation does not refute a formulation intentionally allowing cyclic reuse. General-graph matching gives the classical exact formulation of homogeneous fixed-power pairing. Edmonds [2], Theorem (P), pp. 125–126, supplies its polyhedral foundation through degree and odd-set inequalities. Adding resource incidence and coupled powers creates a different optimization problem.

Denno et al. [9], Sections 2–4, use iterative random sounding and carry the best scheduled subset into the next search round. Their (7) is a regularized inverse-Gram-trace approximation to throughput. The sounding/computation tradeoff is useful practical precedent, while the subset service rule and fixed powers require changes before applying the search to the exact full-service pair rate. Its reported search gains are not a matched benchmark here.

S9.2 Power optimization, decomposition and bounds

Shi et al. [10], Section III, pp. 4022–4025, relate linear-processing rates to MMSE and alternate filters with power allocation. The fixed-filter power problem is solved globally as a geometric program; this does not make the full alternating nonconvex problem globally optimal. Wang et al. [11], Sections II–IV and Algorithm 1, extend iterative weighted-MMSE precoding to multiantenna UEs and APs with Weichselberger fading, channel-error covariance, and centralized or LSFD processing. The scalar-amplitude comparator in Section S12 adapts that mechanism to a common budget and exactly-once service; it does not reproduce the source's full network or figures.

Shen and Yu [12], Theorem 3, give exact auxiliary-variable transformations and stationary-point convergence under the stated concave/convex assumptions. Their Part II [13] exposes weighted bipartite matching in multicell uplink scheduling. Proposition 2 establishes monotonic weighted sum rate and beamformer stationarity when the schedule is fixed. Transformation, matching-based updates and alternating power optimization are established ingredients. The present rectangular pair oracle instead enumerates the complete boundary and interior candidate set for recomputed two-user MMSE rates.

Hassan et al. [14], Theorem 3 and (10), derive a quadratic stationary-candidate calculation for two fixed-gain interfering links after substituting $P_2 = P_T - P_1$. Their detailed theorem proof is referred to

an earlier conference version. The signed-price rectangle studied here has two independent box coordinates and recomputed MMSE coupling. A higher polynomial degree alone is not a new optimization principle. Sharma and Teneketzis [15] study decentralized power allocation with private utilities and externalities under, among other conditions, Assumption 5 of strict concavity. Their MMSE example uses negative MSE utility and a small-correlation approximate-concavity discussion in Appendix B. That utility is different from the logarithmic sum rate, so it does not supply the sharp threshold of main Theorem 2.

Matthiesen et al. [16], Section III, Algorithm 1 and Theorem 1, provide a general mixed-monotone branch-reduce-bound framework converging to a global solution at the stated accuracy under a bound-improving selection rule. The choice of representation affects bound tightness and convergence speed. Our spatial bounds specialize that framework to the exact pair utility; curvature and pair pricing are assessed as bound improvements within it. Barnhart et al. [17], Sections 4–6, provide the restricted-master, pricing and branching framework for large-column integer programs. Complete or appropriately bounded pricing is needed to certify the full relaxation; a restricted master alone is insufficient. Neither column generation nor Lagrangian dual repair is introduced by this article.

Zhao and Pottie [3], Theorem 5, express optimal flat-channel spectrum sharing through a concave envelope with $K + 1$ flat subbands. This is a divisible-frequency allocation, whereas our single-slot model selects one integral pair and one power point per resource. The distinction explains the power-mixture example in Section S3. Le and Pfender [18], Theorems 1 and 5–6, establish tractability and hardness results for maximum rainbow matchings in edge-colored graphs. Their color constraint is an abstract analogue of resource incidence, not a finite-dimensional channel-vector construction or the perfect-3DM premise used in our proof. Kelk and Stamoulis [19] study bounded-color matching LPs and stronger relaxations, including bichromatic four-cycle obstructions. Those combinatorial gaps precede our weights realizable by finite-dimensional channel vectors; the separate continuous-power gap requires a different example.

S9.3 What the earlier hardness results establish

Mostafa et al. [20], Theorem 4 and Remark 1, p. 24461, encode three-dimensional matching through channel orthogonality for decorrelator-SIC offloading. The receiver dimension is $L = |X \times M| + 2$. Their pairwise communication/computation optimization followed by a hypergraph binary program is also close decomposition precedent. Admission, delay and energy constraints differ from unweighted full-service no-SIC sum rate. Our restriction concerns fixed $M = 3$, equal-norm rational channels from a fixed finite alphabet and the specified MMSE threshold; changing the receiver and objective does not make it a direct strengthening of their theorem.

Prasad et al. [21], problem (11) and Theorems 1–2, prove LTE-A uplink scheduling hardness and develop submodular, matroid and knapsack approximation methods under codebook, contiguity, control and interference constraints. Their MAC/log-determinant utility and variable group size differ from fixed two-user no-SIC rates. Salaün et al. [22] give a three-dimensional-matching strong-hardness framework for multicarrier NOMA with SIC, individual budgets, zero-power admission and generalized-mean utilities. These results establish broad joint-allocation hardness; they do not automatically establish the present channel-restricted theorem.

Mollanoori and Ghaderi [23], Section II, Problem 1 and Theorem 1, prove full-service scalar SIC scheduling hardness by a two-slot PARTITION reduction. One common receiver, fixed powers and mandatory service are already present, but arbitrary users per slot and input-dependent received powers carry the reduction. Razaviyayn et al. [24], Section III, Theorems 1–2, establish interference-alignment/DoF hardness with three antennas and a tractable feasibility subclass with at most two antennas per node. The construction uses multiple transmitter/receiver pairs and optimized subspaces. Its dimension boundary does not transfer to finite-SNR MMSE pair scheduling. Liu [25], Section III-C, Theorems 3–4, establishes two-antenna single-carrier transceiver hardness for SINR/power objectives in interference channels, with real or complex channel matrices. These predecessors rule out treating fixed antennas, physical channels, full service or three-dimensional orthogonality individually as new mechanisms.

The external combinatorial premise and the channel realization are also distinct. Dyer and Frieze [26], Section 2, p. 141, statement (2.1), state bounded-occurrence perfect planar-3DM hardness and attribute its proof to their planar-3DM paper. Theorem 1 uses that premise and supplies its own channel-vector reduction. The AIM Minimum Rank Group [1], Theorem 3.16, p. 1639, supplies the real three-dimensional

tree-complement representation antecedent; Section S1 provides the additional rational unit-vector and finite-alphabet construction for the bounded forest family. The result establishes hardness at three coordinates, not a proved transition between dimensions two and three.

S9.4 Cell-free processing, channel acquisition and recent designs

Björnson and Sanguinetti [27] define computational and fronthaul scalability and use dynamic cooperation clustering for cell-free combining/precoding. A coherent stacked receiver does not inherit those distributed implementation properties; our nominal optimizer omits fronthaul-capacity constraints. Ammar et al. [28] give distributed downlink scheduling/beamforming at distributed and central units with channel-error leakage and interference. That architecture motivates uncertainty-aware allocation, while its downlink beam-assignment utility differs from the uplink pair rate. Jiang and Schotten [29], Algorithm 1, compare centralized/distributed combining within a common max-min power-control framework. Their receiver rankings under statistical/estimated CSI and a fairness objective need not hold for instantaneous unweighted pair sum rate.

Cheng et al. [30], Theorem 1, establish optimal training length $\tau_p = K$ and exact power-control formulations for specified MRC/ZF weighted-SE objectives under coupled pilot/payload energy budgets. Jang et al. [31], Section III and Algorithm 1, jointly learn FDD pilots, limited feedback and downlink precoding through a WMMSE-inspired architecture. Thus linking acquisition to the communication objective is established. Our separate fixed-estimator CSI experiment in Section S6 tests actual-channel rates within its own finite scheduling family. Yoo et al. [32], Lemma 2 and Algorithm 1, calibrate context-dependent prediction sets using learned importance ratios with explicit distribution/weight-error terms. Marginal predictive coverage does not by itself give a simultaneous decision-safe outage event under adaptive scheduling.

Alkhateeb's DeepMIMO framework [33], Sections II–IV, generates channels from ray tracing, scene geometry, materials and transmitter/receiver locations, with configurable antennas and subcarriers. It supplies the evaluation inputs, not the constructed channels in the hardness proof. Scene selection, noise, normalization and conditional sampling remain part of our experimental specification.

Chen et al. [34], (4), pp. 3315–3316, integrate grouping, subcarrier allocation and hybrid beamforming for beam-squint-aware wideband downlink transmission with limited RF chains. Constraint (4c) requires service at least once across the frame. Algorithms 1–2, p. 3323, alternate assignment and hybrid-beamformer updates; their changing penalties need not preserve monotonicity of the unpenalized sum rate. These model and guarantee details distinguish that design from mandatory once-only uplink pair scheduling.

Daei et al. [35] optimize pilot spacing and power for nonstationary Rician aging channels. In the journal text, Proposition 2, (7)–(8), p. 3765, uses normalized spatial correlation, and Remark 3, (12)–(13), qualifies the autoregressive approximation. The tagged-user deterministic-equivalent SE objective and power constraint are (38)–(39), p. 3768. Algorithm 1, p. 3769, retains heuristic frame enumeration and projected power ascent. Proposition 3, (40)–(41), concerns the fixed-point calculation used to evaluate SE, rather than global convergence of the outer frame/power optimization.

Elyasi and Vosoughi [36] combine correlation-aware multicell MMSE processing, pilot assignment and pilot/data power design. The journal receiver is (9), p. 33; Theorem 3, (16), p. 34, gives a large-system deterministic equivalent, and Algorithms 1–2, pp. 36–37, address pilot assignment. The weighted sum-SE update, (35)–(38), pp. 38–39, uses a high-SNR log-SINR approximation and matrices evaluated once at initial powers. Those restrictions apply to that update, rather than to every result in the study.

Liu et al. [37] study mobile-AP MR downlink control using permutation-aware multi-agent reinforcement learning, graph communication and reward decomposition. Their journal model and mobility/power objective are (1)–(22), pp. 1819–1821; permutation results are (23)–(33), p. 1822, and Algorithms 1–2 appear on pp. 1823 and 1825. The structural permutation properties and empirical policies address a different decision problem from fixed-channel uplink pairing; they are not instance-wise global scheduling certificates.

Mobini et al. [38] use statistical-CSI MMSE-SIC rates, Proposition 1, (17)–(22), p. 905, with per-AP power budgets and fronthaul-aware association. Algorithm 1 and the reported Fritz John stationarity result for fixed association appear on p. 908. Algorithm 2, pp. 908–909, uses a channel-gain association heuristic. The penalty-limit assertion in Proposition 2 and fixed-association stationarity do not establish global joint association/power optimality. The convergence statement is attributed to the source; its cited proof is not independently re-established here.

Fu et al. [39] optimize statistical-CSI downlink LP-RZF regularization and transmit powers in an EDU–ORU cell-free architecture. Their use-and-then-forget SINR, (8), p. 6447, and Proposition 1, (11)–(18), pp. 6447–6448, give an asymptotic deterministic equivalent under the stated large-system and covariance assumptions. LSF-based top-channel association is fixed before the power/regularization problems, (19) and (23). Algorithms 1–2, p. 6449, use Taylor/SCA subproblems with per-EDU budgets; Section IV-C, p. 6450, reports stationary-point convergence. The distributed statistical downlink objective and that convergence scope differ from exact finite-channel, common-receiver pair scheduling and its validated upper bounds.

Ou et al. [40] provide a closer no-SIC uplink design using two polarized transmit ports per user and a $2M$-port BS with all $K \leq M$ users active. Their estimated-channel SINRs include error covariance, (46)–(47), and the finite-blocklength sum-SE problem, (52)–(55), constrains each user's two stream powers by its own budget. Algorithm 1 computes the combiners before its power loop; the power analysis treats these combiners as fixed. The power-scalar simplification in (76)–(78) and the fixed-combiner convergence/stationarity argument around (75) should not be read as global joint receiver/schedule optimization. The present work instead assigns exactly two users to each orthogonal resource under one shared power budget and recomputes the pair receiver at each power point.

S9.5 Reading versions and interpretation

Citations identify verified publications, while technical reading sometimes uses an author manuscript or preprint. The inspected versions for Wang, Shen–Yu, Matthiesen, Hassan, Zhao–Pottie, Prasad, Le–Pfender, Björnson–Sanguinetti, Cheng, Ammar, Jang, Yoo, DeepMIMO and Jiang–Schotten are listed with their identifiers and locators in the metadata-only `SOURCE_EVIDENCE.json`. That record is supplied in `Supplementary_Materials.zip`. Kelk–Stamoulis is the inspected 2019 version despite the stable bibliography key. The fixed-antenna comparisons use Razaviyayn et al.'s arXiv:1009.3481v1 and Liu's author draft dated 29 October 2016. Liu's displayed (14) has an apparent maximization/minimization inconsistency in extraction; Appendix C explicitly treats total-power minimization. No conclusion here relies on treating that as an error in the published paper.

For Daei, Elyasi–Vosoughi, Liu and Mobini, the relevant journal model, algorithm and guarantee passages were compared with arXiv:2401.13368v1, 2411.02061v2, 2412.02581v1 and 2506.14494v1, respectively. The journal locators above supersede earlier equation numbers; whole-document equivalence and independent verification of every external proof are not assumed. Chen's inspected source is the final publisher-formatted author-hosted PDF. Fu and Ou are assessed from the supplied publisher-formatted journal texts. Conference/journal extensions and preprint/publication versions are not counted as separate evidence.

These comparisons explain attribution and applicability; they neither establish worldwide priority nor transfer the sources' measured performance to our model. In the corrective study, full pricing tightens some continuous upper bounds but adds no target certificate or achieved rate over curvature. The ASU improvements over grid are shared by both spatial arms, and scale pricing leaves grid rates unchanged. The CSI and GPU studies likewise support neither learned-channel robustness of the nominal bounds nor end-to-end acceleration.

S9.6 Relationship to our earlier studies

Our earlier work addresses two complementary parts of this design. Hejazi and Sabaei [41] jointly optimize user clustering and base-station assignment in a user-centric cell-free architecture, then train an ensemble model on optimized assignments. Its NOMA/SIC receiver and multi-AP association differ from the common no-SIC receiver studied here. Hejazi et al. [42] refine an LS channel estimate with a CNN trained using GAN-augmented data to reduce pilot overhead. That estimator supplies upstream CSI; the present bounds apply to fixed nominal channels and do not certify rates under its estimation errors.

S10 Data, code and reproduction from saved results

The project repository is private at the time of this revision; public release is pending. The accompanying delivery contains editable main and supplementary sources, vector figures, original and corrective scalar outcomes, stored rational channel instances, candidate bounds with proof partitions, receipt records, and

analysis and verification programs. The evaluation channels use Alkhateeb's DeepMIMO framework [33] and ray-tracing inputs generated with Remcom Wireless InSite; the software platform supplies propagation data, not the constructed channels in the complexity proof. The release omits the raw `Dataset` directory, raw scenario archives and generated pool NPZ files. These external inputs are separate from the study's stored case coefficients, proof partitions, receipt records and scalar scientific outputs. O1_60 can be obtained from the original O1_60 scenario download. ASU uses the legacy ASU_Campus1 scenario download, also identified by the official DeepMIMO migration guide. Regenerating channels requires separately acquiring the relevant scene inputs and following the generator settings; checking the saved case bounds does not require redistributing the raw scenarios. Publisher full texts, author style samples and compiler binaries are also excluded. The release guide identifies the available files and their revision.

S10.1 What can be reproduced without new optimization

The guide separates four operations: regenerating figures from saved scalars; recomputing the fixed statistical summaries; independently checking saved global bounds; and generating scenes or rerunning optimizers. The first three use saved results. They do not reproduce historical timing on another host. The generator and controller are retained to describe how the studies were run, but new scientific runs require the specified environment and a new timing record.

The independent verifier recomputes feasibility, logarithmic objective enclosures and accepted continuous upper bounds from stored case coefficients and complete proof domains. It imports neither the optimization program nor a numerical optimizer. Integer fixed-point series with explicit remainders enclose the logarithms. Original replay covers 128 cells and 590 receipts. Corrective replay covers 128 cells and 700 receipts. An identical case/schedule/proof input may reuse a mathematical result only after each receipt's integrity, candidate, attempt and timing checks. Thus 700 receipts do not imply 700 distinct numerical reconstructions; the corrective replay reused 417 repeated mathematical inputs. Exact arithmetic concerns the stored channel representation, not an unquantized physical field.

S10.2 Environment and study boundaries

The one-AP calculations used Python 3.12.4, NumPy 2.3.4, SciPy 1.16.3 and SymPy 1.14; DeepMIMO generation used Python 3.9 with the recorded DeepMIMOv3 0.2.9 distribution. The supplied generation manifests retain the exact scenario names and parameter arrays; these legacy inputs should not be silently replaced with a newer scene or API. The recorded host is an Intel Core i7-14700HX Windows system with approximately 64 GiB RAM and single-threaded numerical linear algebra. The host was not established to be idle or dedicated. Exact environments and command outcomes are retained in the manifests. The ancillary GPU environment is specified separately in Section S5.

The historical four-AP study, including incomplete pricing and storage failures, remains distinct from the one-AP evaluation. The current verifier does not reconstruct missing historical proof trees. The original all-zero ASU results and the corrective automatic postprocessing failures are also retained. Section S13 reports the successful 92.894-s saved-data reproduction and the increased computational suballocation that it required. No new channel, optimizer run or inference definition was introduced by that recovery.

The metadata-only `SOURCE_EVIDENCE.json` in `Supplementary_Materials.zip` documents the reading versions and technical locators for the 40 previously inspected references, alongside the comparisons in Section S9. It contains source identifiers and locators, without full texts or internal review records. Journal metadata and the technically inspected version are distinguished; a preprint is not assumed identical to its final publication. Source PDFs, complete extracted texts and source-page renders are internal reading materials and are not part of the reproduction delivery.

The two additional author references are documented in `AUTHOR_REFERENCE_UPDATE.json`, which distinguishes the supplied manuscript versions from verified final publication metadata.

The present local artifact is PDP2-CR030-R1. Its additional `revision/` directory contains a metadata-only provenance record, full generator configurations, a 32-case index, complete scientific-file hashes and statistical references. Section S15 gives the statistical definitions and explains how these records identify exact saved inputs without inventing missing VCS or scenario-version labels. The current figure scripts are

`plot_figures.py` and `draw_system.py` at the archive root. This local hash identity is not a claim of a published DOI or public release.

S11 Full proofs of the structural theorems

These derivations expand main Sections IV–V. Equations D1–D9 are supplementary; other numbered equations refer to the main article. The rational representation is in Section S1 and determinant coefficients are in Section S2.

S11.1 Proof of Theorem 1

Proof. Use perfect three-dimensional matching with sets X, Y, Z of equal cardinality n, where every element occurs in at most three triples. Dyer and Frieze [26], Section 2, p. 141, statement (2.1), state NP-completeness for the planar perfect-matching variant in which every element occurs in two or three triples, and attribute its proof to their subsequent planar-3DM paper. Dropping planarity gives the premise needed here. The construction below realizes the reduction with finite-dimensional channel vectors. Create users $X \cup Y$, resources Z, and $K = 2n$. For a fixed resource z, the admissible triples induce a bipartite graph F_z with at most three edges. It is a forest with at most six nonisolated vertices. Add an isolated outsider role.

Supplementary Section S1 constructs a rational unit representation in $\mathbb{R}^3$ whose zero inner products are exactly the edges of this at-most-seven-role forest. Each nonisolated user receives its role vector on resource z; every remaining user receives the outsider vector. Two outsiders have inner product one, while an outsider and an active role have nonzero inner product. Hence a user pair is orthogonal on z exactly when it corresponds to a triple of the original instance.

At unit power, each selected pair has

$$q_{uvs} = 2\ln\left(2 - \frac{|\mathbf{h}_{us}^T \mathbf{h}_{vs}|^2}{2}\right) \le 2\ln 2, \tag{D1}$$

with equality exactly for an orthogonal pair. A full schedule attains $K \ln 2$ exactly when the original instance has a perfect three-dimensional matching. IDEAL-MMSE-PAIR encodes K, S as integers and its channel table as KS indices into the fixed ordered alphabet of Section S1. A certificate lists the two user indices for every resource, using $O(K \log K)$ bits; incidence and range checks enforce exactly-once service. To avoid transcendental comparison, test $\prod_i (1 + \gamma_i) \ge 2^K$. Each factor is rational with constant bit length. Multiplication gives numerator and denominator with $O(K)$ bits, and cross-multiplication against the $(K + 1)$-bit integer 2^K is polynomial-time. Malformed instances and invalid schedules are rejected. This proves NP-membership; the reduction proves NP-hardness. No arbitrary real threshold is part of this language.

Let $b = \min\{|v^T w| : v, w \in \mathcal{A},\ v^T w \ne 0\} > 0$, including $v = w$ for two users assigned the outsider vector. Any no instance loses at least $\delta_0 = -2\ln(1 - b^2/4) > 0$ from the ideal sum. Under continuous powers, rewrite the pair rate as

$$\begin{aligned} q_{uvs} = {} & \ln(1 + p_u) + \ln(1 + p_v) \\ & + 2\ln\left(1 - r_{uvs}\frac{p_u p_v}{(1 + p_u)(1 + p_v)}\right). \end{aligned} \tag{D2}$$

Jensen's inequality bounds the sum of the first terms by $K \ln(1 + \bar{p})$. Every no instance contains a forbidden pair, whose interaction loss is at least

$$\delta_{\mathrm{box}} = -2\ln\left(1 - b^2\left[\frac{p_{\min}}{1 + p_{\min}}\right]^2\right) > 0. \tag{D3}$$

A yes instance reaches the ideal value with the perfect matching and uniform powers $\bar{p}$. A no instance remains at least δ_{box} below it. The reduction writes $KS = 2n^2$ constant-size channel indices in polynomial

time; $P = K\bar{p}$ has polynomial encoding length. The fixed positive gaps are additive, not constant relative gaps. This continuous-power reduction establishes NP-hardness without an NP-membership claim for unrestricted continuous-threshold problems. □

S11.2 Proof of Theorem 2

Proof. Write $a = 1 - r$, $A = 1 + x$ and $B = 1 + y$. Differentiation gives

$$\begin{aligned} q_{xx} &= A^{-2} - \frac{2(1+ay)^2}{d^2}, \\ q_{yy} &= B^{-2} - \frac{2(1+ax)^2}{d^2}, \qquad q_{xy} = -\frac{2r}{d^2}. \end{aligned} \tag{D4}$$

At $x = 0$, the first expression tends to $1 - 2a^2$ as $y \to \infty$. If $a < 1/\sqrt{2}$, it becomes positive at a finite point, and by continuity at a nearby interior point. Thus global concavity is impossible.

For $a \geq 1/\sqrt{2}$, $A(1+ay) = d - ry$ and $ry/d \leq r$ imply $A(1+ay)/d \geq a$; the two diagonal Hessian entries are nonpositive. The determinant numerator is

$$\begin{aligned} N = {} & \left[2A^2(1+ay)^2 - d^2\right] \\ & \cdot \left[2B^2(1+ax)^2 - d^2\right] - 4r^2A^2B^2. \end{aligned} \tag{D5}$$

Here $\det \nabla^2 q_r = N/(A^2B^2d^4)$, with strictly positive denominator. Section S2 lists every polynomial coefficient. They are nonnegative on $a \in [1/\sqrt{2}, 1]$, including the boundary, with strictly positive constant coefficient. Therefore $N > 0$, and both Hessian eigenvalues are negative at every finite point. □

S11.3 Proof of Theorem 3

Proof. Put $a = 1 - r$, $A = 1 + x$, and $k = 1 + ax$, and define

$$C = A(1 - \lambda_1 A), \qquad B = 2aA - k(1 + \lambda_1 A). \tag{D6}$$

The first interior stationarity equation, multiplied by the positive quantity Ad, is $C + By = 0$. If $C = 0$, then $\lambda_1 A = 1$ and $B = -2r \neq 0$; hence $B = C = 0$ is impossible. Every interior candidate therefore satisfies $y = -C/B$. Substitution into the second stationarity equation gives the polynomial

$$\begin{aligned} \Pi(x) = {} & [2k - (1+\lambda_2)A]B^2 \\ & - [k - \lambda_2(A+k)]CB - \lambda_2 k C^2. \end{aligned} \tag{D7}$$

Its degree is at most five, and the algebraic identity $\Pi(-1) = 2r^3 > 0$ shows that it is not identically zero. The value -1 is used only for this identity and is not feasible. Retain real roots with $L_x < x < U_x$, $B(x) \neq 0$, $L_y < -C(x)/B(x) < U_y$, and both original stationarity equations satisfied. Boundary roots are already covered by edges and corners. Multiplicity does not create additional candidate points.

On a horizontal edge $y = t$, set $h = 1 + at$ and $e = 1 + t$. The stationary equation is

$$2h(1+x) - (e + hx) - \lambda_1(1+x)(e+hx) = 0. \tag{D8}$$

This polynomial has degree at most two and cannot vanish identically: its quadratic coefficient is nonzero when $\lambda_1 \neq 0$, and its linear coefficient is $h > 0$ otherwise. Vertical edges are symmetric. Compactness and the first-order conditions on each face complete the enumeration. □

Degeneracies and numerical realization. The algebra applies to signed real prices, including zero; it also includes $r = 1$. At $r = 0$, $q_0 = \ln(1+x) + \ln(1+y)$ is separable, and the clipped scalar solutions in the main text apply. A zero-width interval leaves the other coordinate's edge equation; two zero-width intervals leave one point. In physical coordinates a zero gain eliminates the rate dependence on that coordinate, leaving only its linear price: choose the lower endpoint for a positive price, the upper endpoint for a negative

price, and either for zero. For the nonnegative network multiplier, choosing minimum power is therefore valid even when both gains vanish.

For rational inputs the separately tested algebraic oracle uses exact rational polynomials and SymPy real-root isolation. It repeatedly removes factors common to Π and B (none can represent a feasible interior stationary point), then takes the square-free part. Distinct repeated roots are retained once. Isolating intervals are intersected with the rectangle; $-C/B$ is enclosed only when the denominator interval excludes zero. Interval candidate boxes may conservatively include a nonstationary point, which cannot invalidate an upper bound. Feasible midpoints give lower bounds. Decimal divisions are rounded outward; logarithms, which use correctly rounded nearest evaluation, are widened with adjacent representable values. The implementation tries root widths $10^{-12}, 10^{-24}, 10^{-36}$ and returns only if the enclosing upper minus feasible lower is at most the requested positive tolerance. A denominator still containing zero or a gap unresolved after those attempts raises an arithmetic error; floating roots are never substituted as proof. This is a rational-input implementation contract, not a precision-independent complexity result. The network experiments instead use the independently checked rectangular-partition algorithm in Section S12.

S11.4 Continuous configuration primal and weak duality

Let $\mathcal{R} = [p_{\min}, p_{\max}]^2$ and let μ_{uvs} be a finite nonnegative Borel measure on $\mathcal{R}$ for every $u < v, s$. Its coordinates represent p_u, p_v. The continuous configuration relaxation is

$$\begin{aligned} V_{\mathrm{CLP}} = \sup_{\mu \geq 0} \quad & \sum_{s,u<v} \int_{\mathcal{R}} q_{uvs}(p_u, p_v)\, d\mu_{uvs} \\ \text{s.t.} \quad & \sum_{s} \sum_{\{u,v\} \ni i} \mu_{uvs}(\mathcal{R}) = 1 \quad (i \in \mathcal{K}), \\ & \sum_{u<v} \mu_{uvs}(\mathcal{R}) = 1 \quad (s \in \mathcal{S}), \\ & \sum_{s,u<v} \int_{\mathcal{R}} (p_u + p_v)\, d\mu_{uvs} \leq P. \end{aligned} \tag{D9}$$

An integral feasible schedule places a unit Dirac mass at the selected powers for each selected pair/resource and zero measures elsewhere, proving $V_{\mathrm{C}} \leq V_{\mathrm{CLP}}$. The objective and moments are finite because the domain is compact and the rate continuous. Given unrestricted α_i, β_s and $\lambda \geq 0$ satisfying every pointwise inequality (8), integration and the mass constraints give $V_{\mathrm{CLP}} \leq \sum_i \alpha_i + \sum_s \beta_s + \lambda P$. Proposition 1 constructs such inequalities from complete pricing bounds. This uses weak duality only and makes no strong-duality assumption.

S11.5 Finite-tolerance best-bound refinement

Fix $\varepsilon > 0$, finitely many assignments, and exact arithmetic. A power box intersects the budget-feasible domain if and only if its lower corner satisfies the budget, since every coordinate is bounded below by that corner. Discard boxes failing this test and evaluate the lower corner of each retained box, updating the global incumbent L. Splits preserve coverage of the feasible domain; feasible points on a shared face may be included in both children without harm.

The mixed-monotone bound and its value at the lower corner differ uniformly by a quantity tending to zero with box diameter: all rate expressions are continuous on the compact original box. The same holds for the affine tangent/chord bounds, since their derivatives are bounded there and their singleton limits agree with the rate; zero-width chords use point values. In particular, taking their minimum with the mixed-monotone bound preserves this property. Therefore a single $\delta > 0$ exists across the finite assignment family such that every feasible box of diameter below δ has upper bound at most its lower-corner value plus ε, hence at most $L + \varepsilon$.

Longest-edge midpoint bisection permits only finitely many tree nodes with diameter at least δ. Indeed, along any branch a selected coordinate is halved, and after finitely many selections every coordinate is shorter than $\delta/\sqrt{K}$; the required depth is uniformly bounded in terms of the original widths and δ. Binary branching

and finitely many roots then give finitely many such nodes. Best-bound selection can consequently process only finitely many boxes with upper bound exceeding $L + \varepsilon$. It terminates when no such box remains. This proves finite-tolerance convergence without an additional unverified fairness assumption. Improving L can only discard more boxes. Finite work caps, incomplete initial assignment coverage, finite-precision arithmetic floors and wall-clock interruption do not satisfy the unrestricted-computation premise; they retain valid fallback intervals but need not attain the tolerance.

S12 Matched joint search and independently checked bounds

S12.1 Matching and power updates

The joint-search comparator adapts the receiver/weight/amplitude mechanism of I-WMMSE [11] and the matching mechanism of uplink fractional programming [13] to exactly-once service and the common power budget. It is not a numerical reproduction of either source's broader system model. Let $a = (s, t)$, $t \in \{1, 2\}$, index the $K = 2S$ slots, and let $\pi(a)$ denote the user assigned to a slot. For noise-whitened channel $\mathbf{h}_{ks}$ and amplitude $v_k = \sqrt{p_k}$, define

$$\mathbf{C}_s = \mathbf{I} + \sum_{a \text{ on } s} p_{\pi(a)} \mathbf{h}_{\pi(a)s} \mathbf{h}_{\pi(a)s}^H, \qquad \mathbf{u}_a = \mathbf{C}_s^{-1} \mathbf{h}_{\pi(a)s} v_{\pi(a)}. \tag{M1}$$

Hermitian linear solves compute the receivers. The scalar MSE and positive weight are

$$e_a = 1 - 2v_{\pi(a)} \Re\{\mathbf{u}_a^H \mathbf{h}_{\pi(a)s}\} + \mathbf{u}_a^H \mathbf{C}_s \mathbf{u}_a, \qquad w_a = e_a^{-1}, \tag{M2}$$

The sum rate equals the maximum of $\sum_a (\ln w_a - w_a e_a + 1)$ over receivers and positive weights. At fixed receivers and weights this expression is a lower bound touching the current rate.

At fixed user powers, the score for assigning user k to slot $a = (s, t)$ is

$$2 w_a v_k \Re\{\mathbf{u}_a^H \mathbf{h}_{ks}\} - p_k \sum_{b \text{ on } s} w_b |\mathbf{u}_b^H \mathbf{h}_{ks}|^2. \tag{M3}$$

A perfect $K \times K$ assignment fills every slot and serves every user once, including when all scores are negative. Receivers and weights are then recomputed. For fixed assignment the amplitude update is

$$\begin{aligned} v_k(\lambda) &= \left[\frac{B_k}{A_k + \lambda} \right]_{\sqrt{p_{\min}}}^{\sqrt{p_{\max}}}, \\ A_k &= \sum_{a \text{ on } s(k)} w_a |\mathbf{u}_a^H \mathbf{h}_{ks(k)}|^2, \quad B_k = w_{a(k)} \Re\{\mathbf{u}_{a(k)}^H \mathbf{h}_{ks(k)}\}. \end{aligned} \tag{M4}$$

The implementation maps zero power-update coefficients to the minimum amplitude. It finds a common nonnegative multiplier using at most 80 bisections, retaining the feasible high endpoint when the sum-power constraint is active. The bracket operator clips to the displayed interval. Every measured decrease in recomputed rate, including roundoff-sized decreases, is rejected. A direct rate check and exact rational power repair retain a feasible incumbent. The deterministic start fills sequential slots with power $\min(p_{\max}, P/K)$. Each of eight seeded starts permutes the users and draws independent residual powers uniformly in $[0, p_{\max} - p_{\min}]$; residuals are scaled down together if their sum exceeds $P - K p_{\min}$, then $p_{\min}$ is added. Exact rational repair enforces feasibility. Each start allows at most 128 outer iterations, stopping when the rate increase is at most $10^{-10} + 10^{-9} \max(1, |f_{\text{previous}}|)$. At most two sweeps of two-resource exchanges follow, with sixteen power refinements per candidate exchange. These finite iteration limits are engineering choices; they imply neither global optimality nor dominance over all recent learning or optimization methods. The actual channel vectors are retained because pair gains and correlations alone do not reconstruct candidate reassignment receivers.

The equation-to-code map is `revision/MATCHING_EQUATION_CODE_MAP.md`; the unchanged implementation is `reproduction/code/corrected/matching_baseline.py`.

Table S3: Algorithm S1: Implemented Matching–WMMSE comparator.

1	Initialize sequential slots with uniform feasible power; generate eight additional starts using `SeedSequence([caller_seed,j])`, $j = 0, \ldots, 7$. Caller seeds are $1101 + i$ for O1 and $1201 + i$ for ASU case index i.
2	For each start, allow at most 128 iterations: compute receivers/weights by (M1)–(M2); maximize the complete user-slot scores (M3) by a perfect assignment; recompute receivers/weights; update all amplitudes by (M4). Negative assignment scores remain eligible.
3	Convert powers to exact rational binary64 values, clip and repair total power by deterministic largest-slack reductions. Reject every rate decrease; stop when improvement is at most $10^{-10} + 10^{-9} \max(1, \lvert R_{\mathrm{old}} \rvert)$ or the search deadline is reached. Retain the best feasible start.
4	Perform at most two two-resource exchange sweeps. Enumerate the six allocations of four users into two labelled pairs, excluding the current allocation, with at most sixteen all-user power refinements per candidate; retain only measured improvements.
5	Stop search at the charged deadline minus five seconds; publish and check retained candidates within the charged deadline, using the reserved interval. A valid incumbent does not imply a global certificate or normal worker completion.

S12.2 Finite work and time limits

The grid stage is capped at 3 s, with separate native ILP and LP caps of 1.25 s inside the common allowance. Search stops 5 s before the charged deadline to leave time for verification and persistence. Pricing and spatial refinement have separate limits of 1,200 splits each. Full allocates at most one quarter of the remaining continuation time to pricing. Spatial search allows at most 3,000 assignment roots; there are $8!/2^4 = 2,520$ at $K = 8$. Unvisited roots retain the universal ceiling. The $K = 16$ grid-plus-pricing arm does not enumerate the scheduling family. These are finite work limits, not convergence criteria or an equal-work comparison.

S12.3 Construction and independent verification of bounds

The optimization program constructs candidate schedules and complete binary box partitions using high-precision logarithm approximations. A separate verifier checks the resulting bounds without importing that program or a numerical optimizer. It reconstructs the pair coefficients as exact rationals from stored binary64 channel entries and checks user coverage, resource occupancy, power bounds and total power. Logarithm enclosures use a 100-bit integer fixed-point atanh expansion with forty terms and an explicit positive tail bound. Every partition split is checked, every terminal region contributes a valid bound, and unlisted spatial roots retain a universal bound. The repaired dual can improve the initial universal bound only after every pair/resource rectangle has been priced.

The controller publishes each candidate and receipt under a unique name after writing, flushing and closing a temporary file. No reader-visible published file is replaced. Only a completed independent verification and completed parent receipt publication by the charged deadline can improve the reported interval. Bounds from different timely receipts for the same case may be intersected. A native solver error is recorded even when an initial valid bound remains available; a return at the time limit is distinguished from a numerical failure. The primary endpoint additionally requires normal worker return by the external deadline.

Offline replay checks all case, candidate, receipt and attempt hashes and parent timing records before accepting any bound. The corrected replay reuses a result only for an identical case hash and identical canonical schedule/proof hash that it has already reconstructed. This offline reuse does not change the timed controller, which clears verifier caches before every charged cell. The checks establish bounds for the stored nominal coefficients, not quantization-free physical channels or unmodeled CSI errors.

S12.4 Exact serialization, arithmetic and rejection rules

Case binding uses the SHA-256 of UTF-8 JSON with sorted keys and compact comma/colon separators. Powers, gains and certified bounds are serialized as rational strings; stored binary64 real/imaginary channel values are reconstructed exactly with integer ratios. The checker does not certify the pre-quantization physical field. It derives the complete partitions itself from rational split events and rejects duplicate/missing

rows, illegal splits, inconsistent channels, infeasibility and unsupported fields. Missing spatial roots require explicit universal fallback coverage; incomplete pair pricing cannot produce a repaired-dual bound.

For a positive rational $z = 2^k q$ with $1 \le q < 2$, put $t = (q-1)/(q+1) \in [0, 1/3)$. The independent checker uses

$$\ln q = 2\sum_{j=0}^{39} \frac{t^{2j+1}}{2j+1} + \mathcal{R}, \quad 0 \le \mathcal{R} \le \frac{2(1/3)^{81}}{81(1-1/9)}. \tag{M5}$$

Each operation is rounded outward by integer floor/ceiling on the 2^{-100} grid; $\ln 2$ is enclosed similarly. Negative range exponents interchange endpoints. Invalid evidence or a verifier deadline raises an exception; an earlier verified fallback remains the only usable bound. Mathematical enclosure, timely parent publication, target attainment and normal worker return are separate statuses. The finite-candidate oracle uses its separate rational-root and directed-Decimal procedure, not this network partition checker. The detailed API and trust boundary are in `revision/CERTIFICATE_API_AND_TRUST.md`.

From the extracted supplementary archive, run

```
python reproduction/reproduce.py -study corrected -mode certificates -output replay
```

using a fresh output directory. The README gives the required environment and the separate statistics/figure modes. This command checks saved proofs and receipts; it neither reruns optimization nor reproduces historical timing.

S13 Original and corrective prospective studies

The original study and the corrective study used separate prespecified protocols, identified as CR023 and CR024 in the repository. Their samples, primary results and errors are reported separately and are not pooled. The original study uses a 32-tone, 50-MHz waveform and a $0.64\,\mu$s useful-symbol window. In its selected ASU rows, 5,694 users have 53,786 retained paths, all later than that window; 2,526 users have no retained paths. The installed generator removes all power beyond the window. Its minimum retained delay is $0.817592\,\mu$s and its maximum is $2.60087\,\mu$s. Consequently all 8,220 generated ASU channels are zero. This is an input/model mismatch, not evidence of physical lack of coverage. All original cases remain in their prespecified analysis.

Table S4: Physical settings of the corrective one-AP study. Array shapes follow DeepMIMO's three-coordinate convention.

Parameter	Setting
Scene / carrier	O1_60 / 60 GHz; ASU_Campus1 / 3.5 GHz
Active access point	AP 1 in each scene
AP / user array	$[1, 8, 1]$ / $[1, 1, 1]$; isotropic elements
AP spacing / rotation	0.5 wavelength / $[0, 0, 0]$ degrees
AP field of view	$[360, 180]$ degrees
Paths / additional effects	Up to ten paths; Doppler and dual polarization disabled
Generator user rows	O1: 321–340; ASU: 21–40
Transform / stored tones	512 over 50 MHz / $0, 16, \ldots, 496$
Main / scale FFT indices	$0, 128, 256, 384$ / $0, 64, \ldots, 448$
Tone spacing / assumed CP	97.65625 kHz / 512 samples ($10.24\,\mu$s)
Noise / noise figure	-174 dBm/Hz / 7 dB; 1.94850×10^{-15} W per tone
Power box / total budget	$[0.05, 0.2]$ W / $K/10$ W
Cases per scene	Twelve with $(K, S) = (8, 4)$; four with $(16, 8)$

The corrective model was fixed before its outcomes were observed: 512 FFT subcarriers over 50 MHz, with 32 stored tones indexed $0, 16, \ldots, 496$. The useful symbol and the assumed conservative 512-sample cyclic guard are each $10.24\,\mu$s. Ideal synchronization, zero carrier-frequency offset, block-static channels and no inter-resource interference are assumed; no time-domain finite-guard receiver is simulated. Every retained absolute delay must be nonnegative and below this common guard/window. No per-user delay shift is applied. A guard or input-validity violation invalidates the complete scene, without resampling.

The channel correction is $\sqrt{512}$ and thermal noise per physical tone is 1.94850×10^{-15} W, using -174 dBm/Hz and a 7-dB noise figure. The stored 32 tones do not change the physical spacing, $50\,\mathrm{MHz}/512 = 97.65625$ kHz. Main resource positions $0, 8, 16, 24$ correspond to actual FFT indices $0, 128, 256, 384$; scale positions $0, 4, \ldots, 28$ correspond to $0, 64, \ldots, 448$. The reported $Q/S = L/(S \ln 2)$ is gross SE normalized by the sum of occupied-tone bandwidths. It excludes the guard and other overheads; the deliberately equal-length guard alone would halve payload SE. These choices describe a generic ideal OFDM benchmark, not a proposed standardized 6G framing or a cross-study algorithmic SNR gain.

The corrective pools use generator rows 321–340 in O1 and 21–40 in ASU. Their coordinates must be disjoint from the original study pools and the earlier O1 development, CSI and confirmation pools. Each network draws distinct users, with replacement across networks conditional on a fixed scene. Zero and weak users remain included. Each scene contributes twelve $K = 8, S = 4$ networks and four $K = 16, S = 8$ networks. Main user seeds are 202611310/311 and scale seeds are 202611320/321. Method-order seed is 202611330; the eight separately specified analytic execution checks use order seed 202611350. The original eight synthetic mechanism controls are reused as existing evidence, not repeated or counted as new samples.

Main cells receive 18 s charged and 20 s external; scale cells receive 28/30 s; analytic execution checks receive 8/10 s. All algorithms, grid levels, power domains, receiver assumptions and split/root limits otherwise remain fixed. The correction retains the original absolute 3,600-s study deadline rather than starting a fresh allowance. The original Python 3.12 controller owns this clock; the Python 3.9 generator receives only relative watchdog durations. A complete method block is reserved before dispatch, with incomplete blocks explicitly missing. The original design reserves 120 s in aggregate for analysis across the original and corrective work. After accounting for the original study's postprocessing, the corrective amendment allocates at most 60 s to its combined automatic analysis/replay block; within that block, the analysis subprocess has a separate 5-s cap. No scientific cell is replaced after its outcome.

Within each study the primary comparison is the equal-scene difference in operational fractions attaining gap at most 0.01 nat. The two discordant probabilities in each of two scene strata receive 98.75% Clopper–Pearson intervals. Bonferroni combination gives at least 95% coverage for the reported difference interval. This is a conservative, resource-sized study, not a powered guarantee of superiority. Other rate, gap, fairness, power, time and memory comparisons remain descriptive or explicitly exploratory. Neither a zero observed difference nor a confidence interval containing zero proves equivalence.

S13.1 Saved-data postprocessing deviation

All 128 scientific cells finished before postprocessing. The automatic analysis subprocess exceeded its 5-s cap without producing a result. The subsequent proof replay reached its remaining deadline, leaving the combined automatic block incomplete after 59.302 s of postprocessing. Both failures and their original files are retained. Amendment CR025 then allocated up to 120 s from the previously unused 180-s overall reserve to one saved-data reproduction attempt using unchanged analysis and verification code on a hash-checked copy. This was an additional allowance: it increased the aggregate analysis allocation beyond the original 120-s reservation shared by the original and corrective work, rather than claiming retrospective compliance with that reservation. The saved-data attempt, including copying, analysis and independent proof replay, succeeded in 92.894 s and remained within the original absolute 3,600-s deadline. No scientific cell was rerun; no new channel, solver run, sample, endpoint or inference definition was introduced. The 700 accepted receipts across 128 corrective cells were independently checked, with 417 repeated mathematical inputs reused only after per-receipt integrity and timing checks.

S14 Original and corrective one-AP results

The two one-AP studies in Section S13 are separate from the historical four-AP experiment in Sections S7–S8. The saved tables include every attempted cell, its status, normal-return flag, eligible bounds, errors, duration, power, fairness and proof coverage. Repeated analytic checks test the implementation; they are not independent propagation samples.

S14.1 Original study and the ASU input mismatch

The original study has 128 normal returns and 590 timely receipts, all independently reconstructed. O1 has twelve joint failures for the full/curvature primary comparison. The all-zero ASU scene has twelve trivial joint successes. Its equal-scene primary difference is zero with conservative 95% interval $[-0.3449, 0.3449]$. These successes are not evidence of useful nondegenerate certification. All 8,220 ASU pool channels were zero after the 0.64-μs generator window clipped every retained path; 5,694 users had retained paths and 2,526 had none. Among the selected rows' retained paths the minimum and maximum delays were 0.817592 and 2.60087 μs. The saved original result is not repaired or pooled with the correction.

Original O1 full and curvature have mean gaps 0.569520 and 0.572798 nat; full and grid have identical main rates. The original four O1 scale cases complete all 960 prices and reduce the mean grid gap from 40.43495 to 7.06737 nat, without changing achieved rates. The degenerate ASU outcomes remain in the CSV and are excluded from any nondegenerate physical interpretation, not from the original prespecified analysis.

S14.2 Corrective study: all executions and scene means

All 96 main, 24 scale and eight analytic reliability executions return normally. No main or scale arm meets the target. Curvature and full each certify both repetitions of the analytic fixture; grid and joint do not. The primary paired table in each scene is $n_{00} = 12, n_{01} = n_{10} = n_{11} = 0$. The resulting difference and interval equal the original numerical values but describe a different, nondegenerate population.

Table S5 reports unpooled scene means; it is descriptive. The complete 128-row export is supplied in the saved-data archive. An achieved SE is $L/(S \ln 2)$, not the natural-log objective; an absolute gap is $U - L$ in nat. Numerical grid ILP/LP values and solver statuses are preserved in each candidate record. Exact feasibility and continuous upper validation remain independent of a solver's optimality status.

Table S5: Corrective means; $n = 12$ per main arm/scene, $n = 4$ per scale arm/scene.

Panel	Scene	Arm	SE	Gap (nat)	Time (s)	Jain
Main	O1	Curvature	27.33527	0.61657	3.680	0.9947
Main	O1	Full	27.33527	0.59237	5.198	0.9947
Main	O1	Grid	27.33527	19.78376	0.584	0.9947
Main	O1	Matching–WMMSE	27.18806	20.19191	2.605	0.9940
Main	ASU	Curvature	11.44830	0.63401	3.814	0.5840
Main	ASU	Full	11.44830	0.54735	5.340	0.5840
Main	ASU	Grid	11.42849	20.96990	0.581	0.5694
Main	ASU	Matching–WMMSE	11.34300	21.20696	2.928	0.5930
Scale	O1	Grid	27.82688	38.86266	0.999	0.9951
Scale	O1	Matching–WMMSE	27.49514	40.70220	5.907	0.9941
Scale	O1	Grid+pricing	27.82688	7.22491	3.412	0.9951
Scale	ASU	Grid	9.59812	38.58020	0.930	0.5957
Scale	ASU	Matching–WMMSE	9.42257	39.55367	8.634	0.6185
Scale	ASU	Grid+pricing	9.59812	3.93320	3.290	0.5957

All main solutions use 0.8 W and all scale solutions use 1.6 W. Full and curvature rates coincide in all 24 main cases. Shared refinement improves on grid in ten ASU cases and ties in two, while all twelve O1 cases tie. Full has tighter upper bounds than curvature in two O1 and nine ASU cases; the other thirteen tie. Its range of main gaps is 0.404865–0.736759 nat in O1 and 0.325381–0.818872 nat in ASU. Scale pricing covers 960/960 rectangles in all eight cases, but gaps remain 7.150558–7.444041 nat in O1 and 3.373034–4.914289 nat in ASU.

S14.3 Where the curvature condition applies

For each saved case, we counted all candidate pair/resource combinations satisfying $r \le 1 - 1/\sqrt{2}$, using the exact equivalent test $2(1-r)^2 \ge 1$. Averaging the case fractions gives 1.12% (O1) and 1.79% (ASU) for the main cases, and 0.94% and 11.64%, respectively, for the scale cases. The denominators include all candidates, not only selected pairs. These descriptive summaries use the existing case records and do not constitute new channel trials or explain algorithmic performance by themselves.

S14.4 Controlled mechanism and isolated oracle diagnostics

The eight synthetic cases in main Fig. 7 have $K = 4, S = 2, M = 4$ and channels

$$\mathbf{h}_{us} = \sqrt{G/4}\big(\sqrt{c}\,\mathbf{z}_{cs} + \sqrt{1-c}\,\mathbf{z}_{us}\big).$$

All underlying vectors are independent $\mathcal{CN}(\mathbf{0}, \mathbf{I}_4)$ draws, except that $\mathbf{z}_{cs}$ is shared by the users on resource s. Resources are independent. The mixture parameter is $c \in \{0, 0.85\}$, the gain scale is $G \in \{1, 100\}$, and seeds are 202611240/241. The parameter c is not the realized squared correlation r; $c = 0$ gives independent channels, not exactly orthogonal channels. Table S6 maps plot order to the saved cases and verified continuation gaps. Each arm starts with the same grid incumbent and numerical dual. A 2-s search allowance precedes independent checking; the recorded total diagnostic duration includes that extra check and may exceed 2 s. These are mechanism controls, not charged-network speed comparisons. Plot positions 1–8 follow the table order rather than a continuous parameter sweep.

Table S6: Original synthetic controls, reused once. Gaps are in nat; no display floor is applied to this table.

ID	c	G	Seed	Reference	Curvature	Full
1	0.0	1.0	202611240	5.25e-28	5.20e-28	5.17e-28
2	0.0	1.0	202611241	1.42e-09	5.41e-15	8.66e-14
3	0.0	100.0	202611240	2.57e-04	3.45e-05	3.45e-05
4	0.0	100.0	202611241	1.43e-04	1.04e-09	1.04e-09
5	0.85	1.0	202611240	5.15e-28	5.15e-28	5.15e-28
6	0.85	1.0	202611241	5.10e-28	5.10e-28	5.10e-28
7	0.85	100.0	202611240	1.67e-06	1.67e-06	1.67e-06
8	0.85	100.0	202611241	2.69e-03	2.69e-03	2.69e-03

All eight diagnostic continuations meet the target. Eligible tangents tighten selected cases with low alignment and high gain; full's repaired dual never beats its final spatial bound in these eight cases. The isolated oracle uses the first pair on the first resource of each synthetic case, its grid-dual price and the physical power box $[0.05, 0.2]^2$ W. All eight calls return under their 2-s watchdog with validated enclosures narrower than the generic target of 10^{-7} nat. Oracle computation and checked generic pricing use different timing boundaries; these measurements establish neither a fair speedup nor a network-level benefit.

S14.5 Solved finite-grid ILP and LP instances

Table S7 gives both solved numerical masters for every corrective network. Status 0 denotes the numerical solver's reported optimum; it is not an exact arithmetic proof or a continuous-domain certificate. The native cap is 1.25 s per solve within the shared method budget. Main masters contain 1,008 configurations and scale masters 8,640. Distinct ILP and LP values show the relaxation gap even when both numerical solves report success.

Table S7: Corrective grid ILP and LP objective values in nat. All 32 networks are retained.

Case	ILP	LP	ILP status	LP status
asu-main-00	32.208132	32.263654	0	0

Case	ILP	LP	ILP status	LP status
asu-main-01	25.040614	25.213387	0	0
asu-main-02	33.415336	33.768411	0	0
asu-main-03	21.062004	21.062004	0	0
asu-main-04	44.934721	44.934721	0	0
asu-main-05	34.580210	34.585543	0	0
asu-main-06	25.822444	25.850471	0	0
asu-main-07	29.243989	29.246271	0	0
asu-main-08	20.467787	20.467787	0	0
asu-main-09	45.079992	45.080084	0	0
asu-main-10	27.334578	27.386179	0	0
asu-main-11	41.048373	41.048373	0	0
asu-scale-00	46.257977	46.257977	0	0
asu-scale-01	72.381528	72.450897	0	0
asu-scale-02	45.699366	45.801507	0	0
asu-scale-03	48.554231	48.554231	0	0
o1-main-00	77.606612	77.636891	0	0
o1-main-01	75.003805	75.039337	0	0
o1-main-02	76.204076	76.204076	0	0
o1-main-03	74.778471	74.778471	0	0
o1-main-04	76.718514	76.718514	0	0
o1-main-05	74.040659	74.040659	0	0
o1-main-06	76.761765	76.761765	0	0
o1-main-07	73.306619	73.306619	0	0
o1-main-08	77.646626	77.646626	0	0
o1-main-09	77.386773	77.386773	0	0
o1-main-10	75.594369	75.594369	0	0
o1-main-11	74.425319	74.425319	0	0
o1-scale-00	155.844316	155.844316	0	0
o1-scale-01	152.390440	152.684282	0	0
o1-scale-02	155.849359	155.849359	0	0
o1-scale-03	153.135755	153.135755	0	0

S15 Revision details: provenance, inference and notation

S15.1 Generator identity and physical normalization

The legacy generator was DeepMIMOv3 0.2.9 with Python 3.9.25 and NumPy 1.26.4. Its installed source files were hashed before the corrective generation. A VCS commit and semantic scenario-release tags were not recorded; `revision/PROVENANCE.json` preserves the installed-source and actual input-file hashes instead. The frozen `construct_deepmimo.py` digest begins `1c0837e2ebffade4`; the full digest and configuration hashes are machine-readable. The original O1 archive digest is unrecorded, while its actual input MAT files are hashed. The ASU archive digest, acquisition URL and all recorded input hashes are retained. No raw archive or pool is redistributed.

The array tuple has X, Y, Z coordinate order; $[1, 8, 1]$ places eight half-wavelength-spaced elements along Y. In `OFDM_PathGenerator.no_LPF`, a path with power P_ℓ, phase ϕ_ℓ and normalized delay n_ℓ contributes

$$\sqrt{P_\ell/N}\exp\{j(\phi_\ell - 2\pi n_\ell k/N)\}, \qquad N = 512. \tag{R1}$$

Multiplication by $\sqrt{N}$ therefore reverses the generator's path-amplitude normalization. It is not an inferred IFFT convention. Generator complex64 outputs are promoted to complex128 and multiplied by $\sqrt{512}/\sqrt{\sigma^2}$, where $\sigma^2 = 1.9484983544617996 \times 10^{-15}$ W. Certification concerns those stored binary64 arrays after this transformation, including prior generator quantization.

The tone spacing is $50\,\text{MHz}/512 = 97.65625$ kHz. The 32 stored tones have aggregate bandwidth 3.125 MHz, but main and scale optimization use only four and eight of them: occupied bandwidths are 0.390625 and 0.78125 MHz. They are spaced across the grid, not contiguous bands. Their gross occupied-band

SE is $L/(S\ln 2)$; the equal useful-symbol/guard durations would multiply payload SE by 1/2 before other overheads. No waveform or timing simulation is implied.

`revision/GENERATOR_CONFIGS.json` contains full parameter arrays, canonical SHA-256 configuration hashes, waveform choices and input-validity records. `revision/CASE_MANIFEST.json` identifies all 32 corrective cases with selection seeds, source pool hashes, original scenario row ranges, selected pool offsets, FFT indices, complete case hashes and channel-array hashes. The selected offsets index the generated pool; they are not physical scenario row IDs. The original per-case JSON and generation manifests remain in `reproduction/studies/corrected/inputs/`.

S15.2 Recorded computational environment

The recorded host is an Intel Core i7-14700HX, 20 physical/28 logical cores, 68,413,153,280 bytes RAM, Windows-11-10.0.26200-SP0. These are retained host records, not a claim of a dedicated or idle host during each cell. Optimization and checking used Python 3.12.4, NumPy 2.3.4, SciPy 1.16.3 and SymPy 1.14.0. The provenance file includes executable and distribution RECORD hashes. Grid optimization uses SciPy `milp` and `linprog(method='highs')`. An independent historical HiGHS version and BLAS implementation/build were not captured. The controller and workers set OpenBLAS, OpenMP and MKL thread variables to one before numerical imports; this is the documented thread policy, not a reconstruction of unrecorded library builds.

S15.3 Exact estimand, interval and resampling construction

Let $Y_{sj}^F, Y_{sj}^C \in \{0,1\}$ be operational target-success statuses for paired case j in scenario s. The registered estimand and estimate are

$$\Delta = \frac{1}{2}\sum_s \{\Pr(Y_s^F = 1) - \Pr(Y_s^C = 1)\}, \qquad \hat{\Delta} = \frac{1}{2}\sum_s \frac{n_{10,s} - n_{01,s}}{n_s}, \quad n_s = 12. \tag{R2}$$

For each of the four discordant probabilities, with count k out of n, the Clopper–Pearson limits [43] are

$$\ell(k,n) = \begin{cases} 0 & k = 0, \\ F^{-1}_{\mathrm{Beta}(k,n-k+1)}(0.00625) & k > 0, \end{cases} \quad u(k,n) = \begin{cases} 1 & k = n, \\ F^{-1}_{\mathrm{Beta}(k+1,n-k)}(0.99375) & k < n. \end{cases} \tag{R3}$$

Each interval has confidence at least 98.75%. A union bound over four intervals gives at least 95% simultaneous coverage [44], with no independence assumption among the four intervals. The scene interval is $[\ell_{10,s} - u_{01,s}, u_{10,s} - \ell_{01,s}]$; the two endpoints are averaged across scenes. If a primary paired status is missing, its scene estimate is undefined and its interval is $[-1, 1]$. All 24 primary pairs are observed here. With zero discordances the overall endpoints are $\pm[1 - 0.00625^{1/12}]$, reported as ± 0.3449.

For a secondary scalar contrast, paired differences d_{sj} are resampled within each scene and averaged with equal scene weights. We report empirical 0.025 and 0.975 quantiles of 10,000 replicate means, following the bootstrap principle [45]. The unchanged code uses `default_rng(202611231)`, consuming the stream first for SE Full–Matching–WMMSE and then gap Full–Curvature, O1 before ASU for each contrast. Only complete valid scalar pairs enter these exploratory intervals; missing values are not zeros. Inference is conditional on the two fixed scenario pools and is not a guarantee across new deployment sites.

S15.4 Execution and receipt accounting

The 128 corrective executions comprise $24 \times 4 = 96$ main-arm runs, $8 \times 3 = 24$ sixteen-user scale runs and $2 \times 4 = 8$ analytic reliability runs on one fixture. The eight original synthetic mechanism cases are separate existing evidence. A receipt is the parent's immutable record of a completed independent check, including its publication time and hashes. The offline replay validated 700 timely receipts; 417 repeated mathematical reconstructions were reused only after each receipt's candidate, case, attempt and timing integrity checks. The cache key combines the exact case hash and canonical schedule/proof hash; timed cells still clear checker caches individually. The 76.4416-s proof replay is part of the broader 92.894-s saved-data recovery, not a replacement value for that disclosed recovery cost.

S15.5 Fairness and minimum-rate summaries

Table S8 reports descriptive means over all twelve main cases per scenario. Jain's index [46] is undefined for an all-zero rate vector; none of the listed main-case indices is missing. These statistics do not change the sum-rate objective or the registered primary endpoint.

Table S8: Mean minimum-user occupied-band SE and Jain index, twelve cases per row.

Scenario	Method	Min. SE (bit/s/Hz)	Jain index
O1	Grid	3.0433	0.9947
O1	Curvature	3.0433	0.9947
O1	Full	3.0433	0.9947
O1	Matching–WMMSE	3.0155	0.9940
ASU	Grid	0.1602	0.5694
ASU	Curvature	0.1530	0.5840
ASU	Full	0.1530	0.5840
ASU	Matching–WMMSE	0.1958	0.5930

S15.6 Compact notation and archive identity

Table S9: Symbols used across the article and methods.

Symbol	Meaning and domain
$K = 2S, M$	User count, resource count and receive dimension.
$\mathcal{K}, \mathcal{S}, \mathcal{E}$	Users, orthogonal resources and unordered user pairs.
$a_i, \mathbf{h}_{is}, \mathbf{n}_s$	Unit-variance transmitted symbol, whitened channel and unit-covariance complex Gaussian noise.
$p_i, p_{\min}, p_{\max}, P$	Transmit powers, individual limits and aggregate radiated-power budget, in W.
g_{is}, r_{uvs}, x, y	Squared channel gain, squared normalized correlation and normalized received-power coordinates.
q_r, q_{uvs}, e_s	Generic pair log-rate, physical-power pair log-rate, and pair selected on resource s.
$Q, Q/S$	Sum of resource SEs in bit/channel-use and gross occupied-band SE in bit/s/Hz.
B_{r}, η	Resource bandwidth in Hz and useful-data fraction.
$z, c, \mathcal{G}$	Binary assignment, pair/resource/power configuration and finite power grid.
$\alpha_i, \beta_s, \lambda$	Unrestricted user/resource prices and nonnegative shared-power price.
$V_{\mathrm{C}}, V_{\mathrm{CLP}}$	Continuous integral-schedule optimum and measure-LP relaxation value, in nat.
L, U	Validated feasible lower value and retained global upper value, in nat.
$U_{\mathrm{dual}}, U_{\mathrm{MM}}, U_0$	Repaired dual bound, mixed-monotone box bound and interference-free fallback, in nat.
$B(x), C(x), \Pi(x)$	Pricing-elimination polynomials; Π is distinct from power budget P.
L_j, U_j, λ_j	Pricing-coordinate interval and price, $j \in \{x, y\}$.
R_i, J	User contribution to occupied-band SE in bit/s/Hz and dimensionless Jain index.

The local artifact identifier is PDP2-CR030-R1. The following digest identifies its scientific-file manifest. `revision/REPRODUCTION_MANIFEST.json`

```
78d8da2a309b2e57dd7f810ae9c7d082
c2339ff17b257643af8aa667992f573b
```

It fixes the complete unchanged scientific payload. Revision additions and delivered ZIP/PDF bytes have separate manifests, avoiding a self-referential archive hash. Public repository visibility and release verification remain outstanding; a local digest is not a published DOI.

References

[1] AIM Minimum Rank–Special Graphs Work Group, "Zero forcing sets and the minimum rank of graphs," *Linear Algebra Appl.*, vol. 428, no. 7, pp. 1628–1648, 2008, doi: 10.1016/j.laa.2007.10.009.

[2] J. Edmonds, "Maximum matching and a polyhedron with 0,1-vertices," *Journal of Research of the National Bureau of Standards, Section B: Mathematics and Mathematical Physics*, vol. 69B, no. 1–2, pp. 125–130, 1965, doi: 10.6028/jres.069b.013.

[3] Y. Zhao and G. J. Pottie, "Optimal Spectrum Management in Multiuser Interference Channels," *IEEE Trans. Inf. Theory*, vol. 59, no. 8, pp. 4961–4976, 2013, doi: 10.1109/tit.2013.2255731.

[4] T. Haustein, "Real time signal processing for multi-antenna systems and experimental verification on a reconfigurable hardware test-bed," Ph.D. dissertation, Technische Universität Berlin, 2006. [Online]. Available: https://edocs.tib.eu/files/e01dd01/522593321.pdf

[5] J. Fan, G. Y. Li, Q. Yin, B. Peng, and X. Zhu, "Joint user pairing and resource allocation for LTE uplink transmission," *IEEE Trans. Wireless Commun.*, vol. 11, no. 8, pp. 2838–2847, 2012, doi: 10.1109/TWC.2012.070912.111305.

[6] M. A. Ruder, D. Ding, U. L. Dang, A. V. Vasilakos, and W. H. Gerstacker, "Joint user grouping and frequency allocation for multiuser SC-FDMA transmission," *Phys. Commun.*, vol. 8, pp. 91–103, 2013, doi: 10.1016/j.phycom.2013.02.004.

[7] M. A. Ruder, A. Moldovan, and W. H. Gerstacker, "SC-FDMA user pairing and frequency allocation with imperfect channel state information," *Phys. Commun.*, vol. 13, pp. 88–98, 2014, doi: 10.1016/j.phycom.2014.10.001.

[8] E. Viterbo and A. Hottinen, "Optimal user pairing for multiuser MIMO," in *Proc. IEEE ISSSTA*, 2008, pp. 242–246, doi: 10.1109/ISSSTA.2008.50.

[9] S. Denno, A. Yamada, and Y. Hou, "Practical uplink multi-user MIMO scheduling based on iterative heuristic search," *IEICE Trans. Commun.*, vol. E108-B, no. 5, pp. 665–672, 2025, doi: 10.23919/transcom.2024EBP3119.

[10] S. Shi, M. Schubert, and H. Boche, "Rate optimization for multiuser MIMO systems with linear processing," *IEEE Trans. Signal Process.*, vol. 56, no. 8, pp. 4020–4030, 2008, doi: 10.1109/TSP.2008.924796.

[11] Z. Wang, J. Zhang, H. Q. Ngo, B. Ai, and M. Debbah, "Uplink Precoding Design for Cell-Free Massive MIMO With Iteratively Weighted MMSE," *IEEE Trans. Commun.*, vol. 71, no. 3, pp. 1646–1664, 2023, doi: 10.1109/tcomm.2023.3235919.

[12] K. Shen and W. Yu, "Fractional Programming for Communication Systems—Part I: Power Control and Beamforming," *IEEE Trans. Signal Process.*, vol. 66, no. 10, pp. 2616–2630, 2018, doi: 10.1109/tsp.2018.2812733.

[13] ——, "Fractional Programming for Communication Systems—Part II: Uplink Scheduling via Matching," *IEEE Trans. Signal Process.*, vol. 66, no. 10, pp. 2631–2644, 2018, doi: 10.1109/tsp.2018.2812748.

[14] N. Ul Hassan, C. Yuen, S. Saeed, and Z. Zhang, "Power Control for Sum-Rate Maximization on Interference Channels Under Sum Power Constraint," *IEEE Trans. Veh. Technol.*, vol. 64, no. 2, pp. 593–609, 2015, doi: 10.1109/tvt.2014.2324289.

[15] S. Sharma and D. Teneketzis, "An externalities-based decentralized optimal power allocation algorithm for wireless networks," *IEEE/ACM Trans. Netw.*, vol. 17, no. 6, pp. 1819–1831, 2009, doi: 10.1109/TNET.2009.2020162.

[16] B. Matthiesen, C. Hellings, E. A. Jorswieck, and W. Utschick, "Mixed Monotonic Programming for Fast Global Optimization," *IEEE Trans. Signal Process.*, vol. 68, pp. 2529–2544, 2020, doi: 10.1109/tsp.2020.2983284.

[17] C. Barnhart, E. L. Johnson, G. L. Nemhauser, M. W. P. Savelsbergh, and P. H. Vance, "Branch-and-price: Column generation for solving huge integer programs," *Oper. Res.*, vol. 46, no. 3, pp. 316–329, 1998, doi: 10.1287/opre.46.3.316.

[18] V. B. Le and F. Pfender, "Complexity results for rainbow matchings," *Theor. Comput. Sci.*, vol. 524, pp. 27–33, 2014, doi: 10.1016/j.tcs.2013.12.013.

[19] S. Kelk and G. Stamoulis, "Integrality gaps for colorful matchings," *Discrete Optim.*, vol. 32, pp. 73–92, 2019, doi: 10.1016/j.disopt.2018.12.003.

[20] S. Mostafa, C. W. Sung, and Y. Guo, "Joint computation and communication resource allocation with NOMA and OMA offloading for multi-server systems in F-RAN," *IEEE Access*, vol. 10, pp. 24 456–24 466, 2022, doi: 10.1109/ACCESS.2022.3152531.

[21] N. Prasad, H. Zhang, H. Zhu, and S. Rangarajan, "Multi-User MIMO Scheduling in the Fourth Generation Cellular Uplink," *IEEE Trans. Wireless Commun.*, vol. 12, no. 9, pp. 4272–4285, 2013, doi: 10.1109/twc.2013.072513.120734.

[22] L. Salaün, C. S. Chen, and M. Coupechoux, "Optimal Joint Subcarrier and Power Allocation in NOMA Is Strongly NP-Hard," in *2018 IEEE International Conference on Communications (ICC)*, 2018, pp. 1–7, doi: 10.1109/icc.2018.8422362.

[23] M. Mollanoori and M. Ghaderi, "On the complexity of wireless uplink scheduling with successive interference cancellation," in *Proc. 49th Allerton Conference on Communication, Control, and Computing*, 2011, pp. 1064–1071, doi: 10.1109/allerton.2011.6120285.

[24] M. Razaviyayn, M. Sanjabi, and Z.-Q. Luo, "Linear Transceiver Design for Interference Alignment: Complexity and Computation," *IEEE Trans. Inf. Theory*, vol. 58, no. 5, pp. 2896–2910, 2012, doi: 10.1109/tit.2012.2184909.

[25] Y.-F. Liu, "Dynamic Spectrum Management: A Complete Complexity Characterization," *IEEE Trans. Inf. Theory*, vol. 63, no. 1, pp. 392–403, 2017, doi: 10.1109/tit.2016.2623662.

[26] M. E. Dyer and A. M. Frieze, "On the complexity of partitioning graphs into connected subgraphs," *Discrete Appl. Math.*, vol. 10, no. 2, pp. 139–153, 1985, doi: 10.1016/0166-218x(85)90008-3.

[27] E. Björnson and L. Sanguinetti, "Scalable Cell-Free Massive MIMO Systems," *IEEE Trans. Commun.*, vol. 68, no. 7, pp. 4247–4261, 2020, doi: 10.1109/tcomm.2020.2987311.

[28] H. A. Ammar, R. Adve, S. Shahbazpanahi, G. Boudreau, and K. V. Srinivas, "Distributed Resource Allocation Optimization for User-Centric Cell-Free MIMO Networks," *IEEE Trans. Wireless Commun.*, vol. 21, no. 5, pp. 3099–3115, 2022, doi: 10.1109/twc.2021.3118303.

[29] W. Jiang and H. D. Schotten, "What is the Most Efficient Technique for Uplink Cell-Free Massive MIMO?" in *2025 IEEE 102nd Vehicular Technology Conference (VTC2025-Fall)*, 2025, pp. 1–6, doi: 10.1109/vtc2025-fall65116.2025.11310428.

[30] H. V. Cheng, E. Björnson, and E. G. Larsson, "Optimal Pilot and Payload Power Control in Single-Cell Massive MIMO Systems," *IEEE Trans. Signal Process.*, vol. 65, no. 9, pp. 2363–2378, 2017, doi: 10.1109/tsp.2016.2641381.

[31] J. Jang, H. Lee, I.-M. Kim, and I. Lee, "Deep Learning for Multi-User MIMO Systems: Joint Design of Pilot, Limited Feedback, and Precoding," *IEEE Trans. Commun.*, vol. 70, no. 11, pp. 7279–7293, 2022, doi: 10.1109/tcomm.2022.3209887.

[32] S. Yoo, S. Park, P. Popovski, J. Kang, and O. Simeone, "Calibrating Wireless AI via Meta-Learned Context-Dependent Conformal Prediction," *IEEE Trans. Signal Process.*, vol. 74, pp. 423–438, 2026, doi: 10.1109/tsp.2026.3650912.

[33] A. Alkhateeb, "DeepMIMO: A generic deep learning dataset for millimeter wave and massive MIMO applications," in *Proc. Information Theory and Applications Workshop (ITA)*, San Diego, CA, USA, Feb. 2019, pp. 1–8. [Online]. Available: https://www.wi-lab.net/research/localization-digital-twins/

[34] K. Chen, C. Qi, and O. A. Dobre, "Integrated User Grouping, Subcarrier Allocation, and Hybrid Beamforming for Wideband Multiuser mmWave Massive MIMO," *IEEE Trans. Commun.*, vol. 74, pp. 3313–3328, 2026, doi: 10.1109/TCOMM.2026.3652527.

[35] S. Daei, G. Fodor, M. Skoglund, and M. Telek, "Toward Optimal Pilot Spacing and Power Control in Multi-Antenna Systems Operating Over Non-Stationary Rician Aging Channels," *IEEE Trans. Commun.*, vol. 73, no. 6, pp. 3761–3777, 2025, doi: 10.1109/TCOMM.2024.3490500.

[36] M. Elyasi and A. Vosoughi, "Massive MIMO Over Correlated Fading Channels: Multi-Cell MMSE Processing, Pilot Assignment and Power Control," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 30–46, 2026, doi: 10.1109/TWC.2025.3581057.

[37] Z. Liu, J. Zhang, Y. Zhu, E. Shi, and B. Ai, "Mobile Cell-Free Massive MIMO With Multi-Agent Reinforcement Learning: A Scalable Framework," *IEEE Trans. Wireless Commun.*, vol. 24, no. 3, pp. 1817–1831, 2025, doi: 10.1109/TWC.2024.3512663.

[38] Z. Mobini, A. H. Gokceoglu, L. Wang, G. Peters, and H. Q. Ngo, "Fronthaul-Aware User-Centric Generalized Cell-Free Massive MIMO Systems," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 900–914, 2026, doi: 10.1109/TWC.2025.3587672.

[39] J. Fu, P. Jiang, P. Zhu, H. Q. Ngo, and M. Matthaiou, "Deterministic equivalent-based resource allocation for cell-free massive MIMO," *IEEE Trans. Commun.*, vol. 74, pp. 6443–6460, 2026, doi: 10.1109/TCOMM.2026.3672179.

[40] X. Ou, S. Dang, and A. Doufexi, "Uplink RSMA with dual-polarized multi-user MIMO over correlated rician channels in finite blocklength regime," *IEEE Trans. Commun.*, vol. 74, pp. 14 516–14 531, 2026, doi: 10.1109/TCOMM.2026.3731763.

[41] P. Hejazi and M. Sabaei, "Optimizing spectral efficiency through user clustering in user-centric cell-free massive MIMO environment for 6G wireless network," *Journal of Communications and Networks*, vol. 28, no. 1, pp. 30–39, 2026, doi: 10.23919/JCN.2025.000085.

[42] P. Hejazi, M. Sabaei, and A. Hajmahmoudi, "Pilot-efficient channel estimation for 6G MU-mMIMO using LS initialization and GAN-augmented CNN refinement," *Journal of Communications and Networks*, vol. 28, no. 3, pp. 401–408, 2026, doi: 10.23919/JCN.2025.000112.

[43] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934.

[44] NIST/SEMATECH, "e-handbook of statistical methods: Bonferroni's method, sec. 7.4.6.3," https://www.itl.nist.gov/div898/handbook/prc/section4/prc463.htm, accessed Nov. 2, 2026.

[45] B. Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.

[46] R. Jain, D.-M. Chiu, and W. R. Hawe, "A quantitative measure of fairness and discrimination for resource allocation in shared computer systems," Digital Equipment Corporation, Tech. Rep. TR-301, Sep. 1984, available as arXiv:cs/9809099 (1998 archival posting).